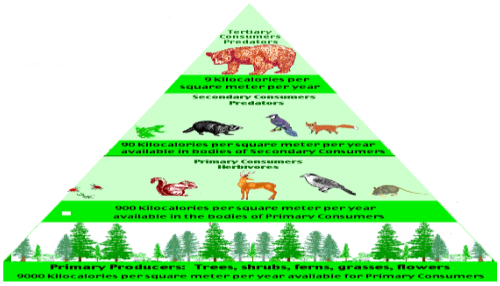
Unit 1 Question bank

Part--- A 2 Marks Questions			
1	**Explain the significance of food web &food chain?**	**Level-2**	**CO1**
Ans.	The transfer of food energy from the producers (plants) through a series of organisms (Herbivores, Carnivores) successively with the repeated activities of eating and being eaten is known as food chain. Food web is a network of food chains where different types of organisms are connected at different trophic levels so that there are a number of options of eating and being eaten at each trophic level Both processes should occur to balance the ecosystem, otherwise species predominating in number and leads to disturbance.		
2	**list the abiotic components of an eco-system with examples?**	**Level-1**	**CO1**
Ans.	Abiotic structure includes the non-living things of the ecosystem such as physical factors (eg: soil, temperature, light & water) and chemical factors consisting the inorganic compounds (eg N, C, H, K, P, S) & organic compounds (g: carbohydrates,proteins).		
3	**Explain any 2 types of components of ecosystems**	**Level-2**	**CO1**
Ans.	The main two components of ecosystem **Abiotic structure** includes the non-living things of the ecosystem such as physical factors (soil, temperature, light & water) and chemical factors consisting the inorganic compounds (N, C, H, K, P, S) & organic compounds (carbohydrates proteins). **Biotic structure** includes plants; animals & microorganisms present in an ecosystem form the biotic component.		
4	**What is Detritus food chain?**	**Level-2**	**CO1**
Ans.	This food chain starts from dead organic matter (dead leaves / plants / animals) and goes to Herbivores and on to Carnivores and so on. A good example of detritus food chain is seen in a Mangrove (estuary). Mangrove leaf fragments acted on by saprotrophs (fungi, bacteria), colonized by algae are eaten by detritus consumers (crabs, shrimps, nematodes, molluscss etc.). These are, in turn, eaten by minnows and small carnivorous fish which serve as the food for large game fish and birds.		
5	**Write the significances of biogeochemical cycle in pollution?**	**Level-2**	**CO1**
Ans.	Biogeochemical cycle involves in circulation of chemicals in between biotic and abiotic by the name of process food chain and food web. Because of continuous moving of chemical, we can avoid the over deposition at one particular place otherwise it leads to pollution.		

<table>
<tr><td>6</td><td>Differences between terrestrial &aquatic ecosystem?</td><td>Level-2</td><td>CO1</td></tr>
<tr><td>Ans.</td><td colspan="3">Any ecosystem can be differentiate basing on structural components
Terrestrial ecosystem:


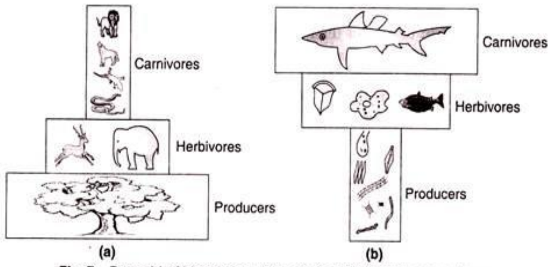
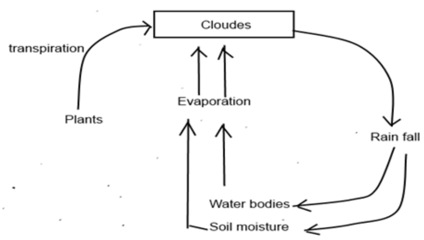


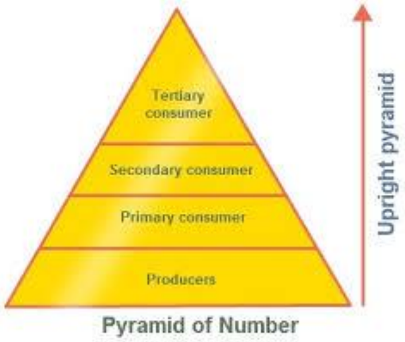
Fig. 7. Pyramid of biomass in (a) Forest and (b) Pond ecosystems.
Eg: Forest structural components Eg: Aquatic ecosystem</td></tr>
<tr><td>7</td><td>Why are green plants regarded as producers?</td><td>Level-2</td><td>CO1</td></tr>
<tr><td>Ans.</td><td colspan="3">Producers are mainly chlorophyll bearing green plants (photo autotrophs) which can synthesize their food in presence of sunlight making use of CO2 and water through the process of photosynthesis. Since plants convert solar energy into chemical energy so they must be better called converters or transducers.</td></tr>
<tr><td>8</td><td>Define the hydrological cycle and explain</td><td>Level-1</td><td>CO1</td></tr>
<tr><td>Ans.</td><td colspan="3">Water cycle or hydrological cycle: - Water from the transpiring plants, oceans, rivers lakes evaporate in the atmosphere.
1. These water vapor subsequently cool and condense to form clouds and water.
2. Water returns to the earth as rain and snow
Diagrammatical representation:

</td></tr>
<tr><td>9</td><td>What is meant by bioaccumulation?</td><td>Level-1</td><td>CO1</td></tr>
<tr><td>Ans.</td><td colspan="3">The accumulation of a substance like pesticide (DOT), methyl mercury of other organic chemicals in an organism or part of an organism (tissue) is called bioaccumulation. Bioaccumulation refers to the gradual build-up of pollutants in living organisms.</td></tr>
<tr><td>10</td><td>Define autotrophs</td><td>Level-1</td><td>CO1</td></tr>
<tr><td>Ans.</td><td colspan="3">The green plants have chlorophyll with the help of which they trap solar energy and change it into chemical energy of carbohydrates using simple inorganic compound namely, water and carbon dioxide. This process is known as photosynthesis. The chemical energy stored by the</td></tr>
</table>

	producers is utilized partly by the producers for their own growth and survival and the remaining is stored in the plants for their future use. **Ex**: plants, photosynthetic bacteria

PART–B 5 Marks Questions			
1	**Explain the structure of ecosystem.**	**Level-2**	**CO1**
Ans.	The term refers to various components. An ecosystem has two Major component. (I) Biotic (living) components (II) Abiotic (non-living) components (I) Biotic components: Living components in an ecosystem are either producers or consumers. They are also called biotic components. Producers can produce organic components e.g. plants can produce starch, carbohydrates, cellulose from a process called photosynthesis. The term refers to various components. An ecosystem has two Major component. (I) Biotic (living) components (II) Abiotic (non-living) components **(I) Biotic components:** Living components in an ecosystem are either producers or consumers. They are also called biotic components. Producers can produce organic components e.g. plants can produce starch, carbohydrates, cellulose from a process called photosynthesis. Consumers are the components that are dependent on producers for their food e.g. human beings and animals Biotic Components are further classified into three main groups 1. Producers 2. Consumers 3. Decomposers or Reducers **1.Producers (Autotroph):** The green plants have chlorophyll with the help of which they trap solar energy and change it into chemical energy of carbohydrates using simple inorganic compound namely, water and carbon dioxide. This process is known as photosynthesis. The chemical energy stored by the producers is utilized partly by the producers for their own growth and survival and the remaining is stored in the plants for their future use. They are classified into two categories based on their source of food. a) Photoautotrophs: An organism capable of synthesizing its own food from inorganic substances using light as an energy source. Green plants and photosynthetic bacteria are photoautotrophs. b) Chemotrophs: Organisms that obtain energy by the oxidation of electron donors in their environments. These molecules can be organic (chemoorganotrophs) or inorganic (chemolithotrophs). **2. Consumers (Heterotrophs)**: The animals lack chlorophyll and are unable to synthesis their own food therefore they depend on the producers for their food. They are known as heterotrophs (i.e. heteros = others, trophs = feeder). The Consumers are of three types:		

	(a) Primary Consumer (Herbivores): i.e. Animal feeding on plants, e.g. Rabbit, deer, goat etc. (b) Secondary Consumers (Primary carnivores): The animal feeding on Herbivores are called as secondary Consumer or primary carnivores. e.g. Cats, foxes, snakes. (c) Tertiary Consumers (secondary carnivores): These are directly depending on the Secondary and primary consumers for their food. e.g. Tiger, Lion etc **3. Decomposers or Reducers:** Bacteria & fungi belong to this category. They break down the dead organic matter of producers & consumers for their food and release to the environment the simple inorganic and organic substance. These simple substances are reused by the producers resulting in a cyclic exchange of material between biotic & abiotic environment. Eg: Bacteria, Earth worms, Beetles etc. **(II) Abiotic components:** The non-living components (physical & chemical) of ecosystem collectively. The abiotic component of the ecosystem is divided into three types ¬ Climatic factors: Solar energy, temperature, air etc. Physical factors: light fire. Chemical factors: Acidity, Salinity, inorganic nutrients Like C, H, O, P, K, N and organic substance like proteins, lipids, carbohydrates etc.		
2	**Describe the ecological pyramids.**	**Level-2**	**CO1**
Ans.	**Ecological pyramids:** The different species in a food chain are called tropic levels. Each food chain has 3 mains trophic level, producer, consumer, and decomposers. Thus, Graphical representation of these trophic levels is called as Ecological Pyramids. It was devised by an ecologist “Charles Elton” therefore this pyramid is also called Ecological pyramid or Eltonian pyramids. Types of Ecological Pyramids: Ecological pyramids are of three types: 1. Pyramid of numbers. 2. Pyramid of energy. 3. Pyramid of biomass. **1. Pyramid of numbers:** It represents the number of individual organisms present in each tropic level. E.g. A grassland Ecosystem Producers are grass (small in size and large in number. Hence, they occupy the first tropic level. The primary consumers are rats occupying the second tropic level. It is worthwhile to note that		

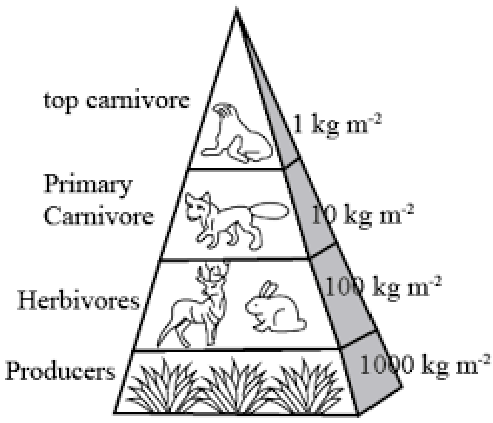
rats are less in number than grass. Secondary consumers are snakes which occupy the third tropic level and they are lesser in number than rats. Tertiary consumers are Eagles that occupy the next tropic level. This is the last tropic level where the number and size of the tropic level is the least as shown in the diagram.



2. Pyramid of energy:

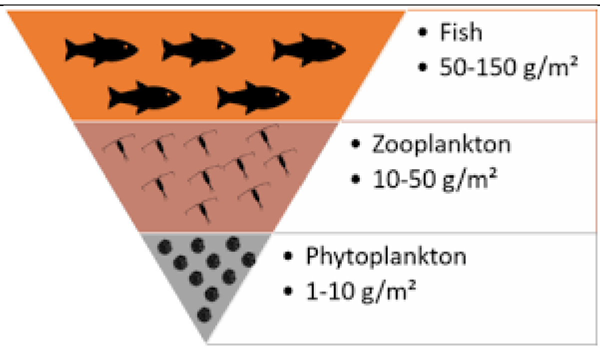
It represents the amount of energy present in each tropic level. The rate of energy flow and the productivity at each successive tropic level is shown in the figure below. At every successive tropic level, there is a heavy loss of energy (almost 90%) in the form of heat.

Thus, at each tropic level only 10% is transferred. Hence there is a sharp decrease in energy at each and every successive tropic level as we move from producers to top consumers (carnivores). It is always upright.



3. Pyramid of biomass:

It represents the total amount of biomass (mass or weight of biological material) present in each tropic level. Considering the example of a forest ecosystem, there is a steady decrease in the biomass from the lower tropic level to the higher tropic level. The producers (trees) contribute a major amount of the biomass. The next tropic levels are the herbivores (insects and birds) and carnivores (snakes, foxes, etc). The top of the tropic level consists of very few tertiary consumers (Ex: Lions and Tigers) whose biomass is very low.

	 		
3	**How the energy flow in the ecosystems**	**Level-3**	**CO1**
Ans.	Biological activities require energy which ultimately comes from the sun. Solar energy is transformed into chemical energy by a process of photosynthesis this energy is stored in plant tissue and then transformed into heat energy during metabolic activities. Thus, in biological world the energy flows from the sun to plants and then to all heterotrophic organisms. The flow of energy is unidirectional and non-cyclic. This one-way flow of energy is governed by laws of thermodynamics which states that: (**a**) Energy can neither be created nor be destroyed but may be transformed from one form to another. (**b**) During the energy transfer there is degradation of energy from a concentrated form (mechanical, chemical, or electrical etc.) to a dispersed form (heat). No energy transformation is 100 % efficient; it is always accompanied by some dispersion or loss of energy in the form heat. Therefore, biological systems including ecosystems must be supplied with energy on a continuous Basis. The flow of energy in an ecosystem follows the laws of thermodynamics. **I law of thermodynamics -** "Energy neither can be created nor destroyed, but it can be converted from one form to other". Energy for an ecosystem comes from the sun. It is absorbed by plants; it is converted into chemical energy. This chemical energy utilized by consumers transform into heat. **II law of thermodynamics -** "Whenever energy is transformed, there is a loss of energy through the release of heat". Energy is transferred between tropic levels in the form of heat as it moves from one tropic level to another tropic level. The loss of energy takes place through work, running, hunting etc., Flow of energy and nutrient cycling from abiotic to biotic and vice versa.		

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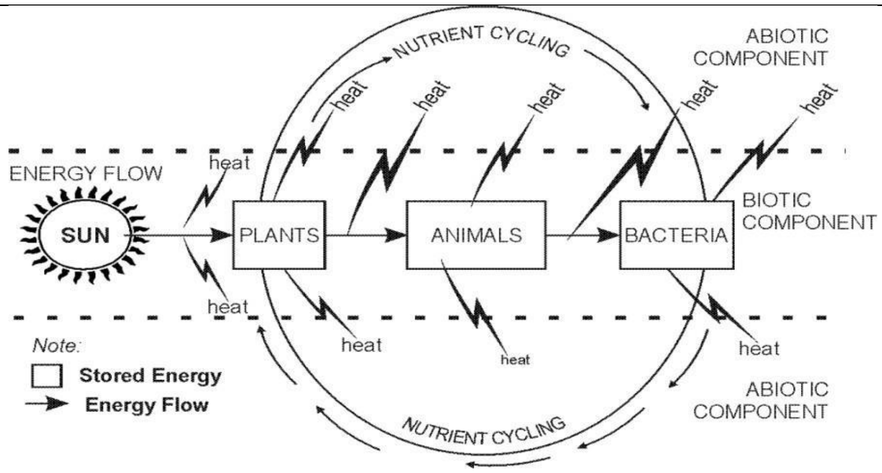
<tr><td></td><td colspan="3">
</td></tr>
<tr><td>4</td><td>Explain the Food chain, food web in ecosystem</td><td>Level-2</td><td>CO1</td></tr>
<tr><td>Ans.</td><td colspan="3">There sequence of eating and being eaten in an ecosystem is known as “food chain” (or) Transfer of food energy from the plants through a series of organisms is known as food chain” A food chain always starts with plant life and ends with animal. When the organisms die, they are all decomposed by microorganism (bacteria and fungi) into nutrients that can again be used by the plants. At each and every level, nearly 80-90% of the potential energy gets lost as heat
Green plants → Deer → Tiger (or) lion
Types of Food Chains: There are two types of food chains
1. Grazing food chain
2. Detritus food chain
1. Grazing food chain: Found in Grassland ecosystems and pond ecosystems. Grazing food chain starts with green plants (primary producers) and goes to decomposer food chain or detritus food chain through herbivores and carnivores.
2. Detritus food chain: Found in Grassland ecosystems and forest ecosystems. Detritus food chain starts with dead organic matter (plants and animals) and goes to decomposer food chain through herbivores and carnivores.
Dead bodies of animals
(Decomposed by bacteria) ⟶ Flies ⟶ Frog ⟶ snake ⟶ Hawk
Food web
The interlocking pattern of various food chains in an ecosystem is known as food web. In a food web many food chains are interconnected, where different types of organisms are connected at different tropic levels, so that there are a number of opportunities of eating and being eaten at each tropic level.
Grass may be eaten by rabbit, rats, deer's, etc., these may be eaten by carnivores (snake, fox, tiger). Thus, there is an interlocking of various food chains called food webs.</td></tr>
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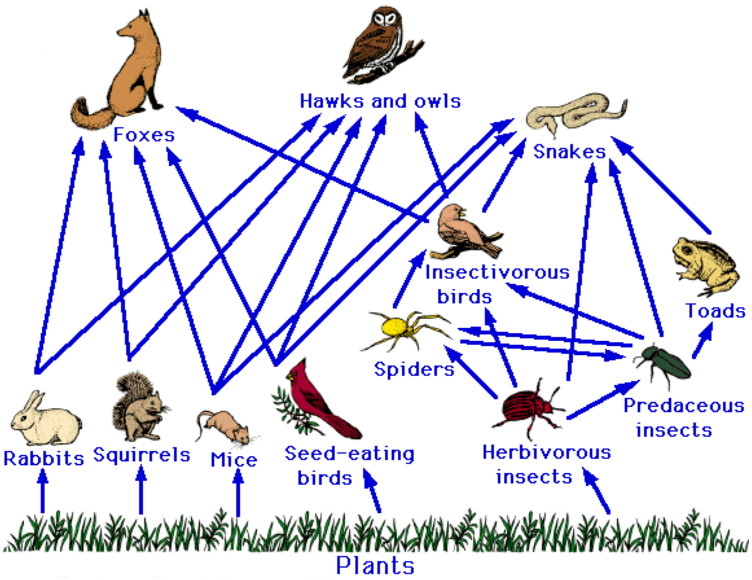

Significance of food chains and food webs

Food chains and food webs play a very important role in the ecosystem. Energy flow and nutrient cycling takes place through them.
They maintain and regulate the population size of different tropic levels, and thus help in maintaining ecological balance.
They have the property of bio-magnification. The non – biodegradable materials keep on passing from one tropic level to another.
At each successive tropic level, the concentration keeps on increasing.
This process is known as bio-magnification

</td></tr>
<tr><td>5</td><td>Write the classification of ecosystem</td><td>Level-2</td><td>CO1</td></tr>
<tr><td>Ans.</td><td colspan="3">Due to the abiotic factors, different ecosystems develop in different ways. These factors and their interaction between each other and with biotic components have resulted in formation of different types of ecosystems as explained below. Ecosystem may be natural or artificial.
Artificial Ecosystem: These are maintained or created artificially by man. The man tries to control biotic community as well as physico-chemical environment. Eg: Artificial Pond, urban area development.
Natural Ecosystem:
It consists of Terrestrial and Aquatic Ecosystems which are maintained naturally.</td></tr>
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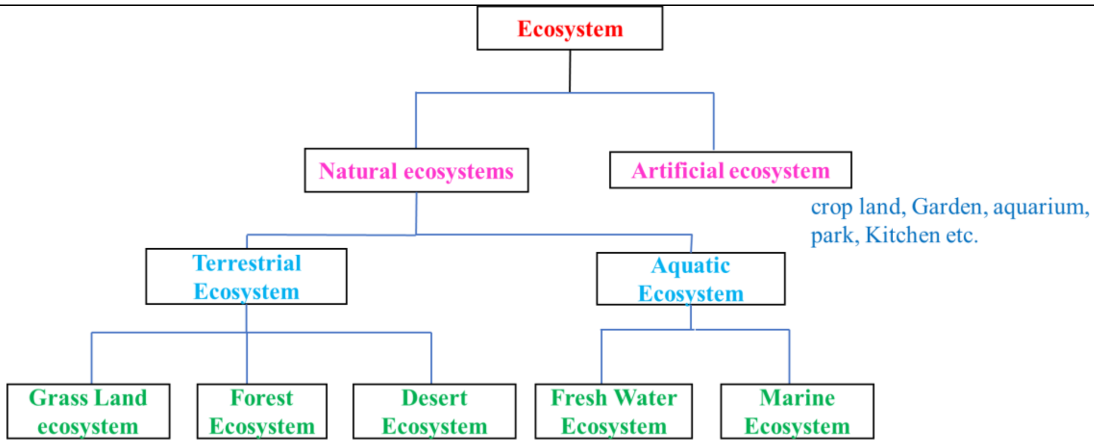


Terrestrial Ecosystem

Terrestrial ecosystems are many because there are so many different sorts of places on Earth. Some of the most common terrestrial ecosystems that are found are the following:

1. Grassland ecosystem
2. Forest ecosystem
3. Desert ecosystem

1. Grass land ecosystem:

These occupy a comparatively fewer area, roughly 20% of the earth's surface.

The various components of grassland ecosystem are:

Abiotic components: These include the nutrients present in soil and the aerial environment. C, H, O, N, P, S are supplied by Carbon dioxide, water, nitrates, phosphates and sulphates.

Biotic components: These may be categorized as:

Producers: These are mainly grasses e.g. Cynodont species, Dicanthium species etc. Shrubs may also be present.

Consumers: These occur in the following sequence:

Primary consumers are the herbivores feeding on grasses are mainly grazing animals as cows, deer's and rabbit etc. Besides them there are insects, termites and millipedes that feed on leaves.

Secondary consumers: Carnivores feeding on herbivores **e.g.** Fox, Snakes, frogs, Lizards etc.

Decomposers: Microbes including fungi like Mucor, Aspergillus, Rhizopus etc and some bacteria and actinomycetes.

2.Forest ecosystem:

Forests occupy roughly 40 % of the land. In India these occupy roughly one-tenth of the total land area. The different components of forest ecosystem include:

Abiotic components: Include inorganic and organic substances present in soil and atmosphere. Dead organic debris is also present in forests.

Biotic components: These include producers like trees, shrubs and ground vegetation. Among the primary consumers are the herbivores which include animals feeding on tree leaves as ants, flies, beetles, spiders etc. and larger animals like elephants, deer etc. Secondary consumers are the carnivores like snakes, birds and fox etc. Tertiary consumers are the top carnivores like lion, tiger etc. Decomposers include microorganisms like fungi, bacteria and actinomycetes

3.Desert ecosystem: Desert occupy about 17% of land, occurring in the regions with an annual rainfall of about 25 centimeters. The species composition of such an ecosystem is varied and typical due to the extremes of temperature and water factors.

The biotic components include:

- Producers are shrubs, especially bushes, some grasses and a few trees.
- Consumers include animals like reptiles and insects which are able to live under xeric conditions. In addition, there are some nocturnal rodents and birds. The camel feeds on tender shoots of the plants.
- Decomposers are very few, as due to poor vegetation the amount of dead organic matter is correspondingly less. They are some fungi and bacteria

Aquatic ecosystem:

An aquatic ecosystem is an ecosystem that exists in water. Within an aquatic ecosystem, the environment is a watery one. Aquatic ecosystems can be divided into freshwater ecosystems (such as fresh rivers or freshwater lakes) and marine ecosystems such as the sea and rock pools.

1. Ponds ecosystem:

A Pond as a whole serves a good example of freshwater ecosystem.

Abiotic Components: The chief components are heat, light, PH of water, CO2, oxygen, calcium, nitrogen, phosphates, etc.

Biotic Components: The various organization that constitute the biotic component are as follows,

Producers: These are green plants, and some photosynthetic bacteria. The producer fix radiant energy and convert it into organic substances as carbohydrates, protein etc.

Macrophytes: these are large rooted plants, which include partly or completely submerged hydrophytes, eg Hydrilla, Trapha, Typha.

P**hytoplankton**: These are minute floating or submerged lower plants eg algae.

Consumers: They are heterotrophs which depend for their nutrition on the organic food manufactured by producers.

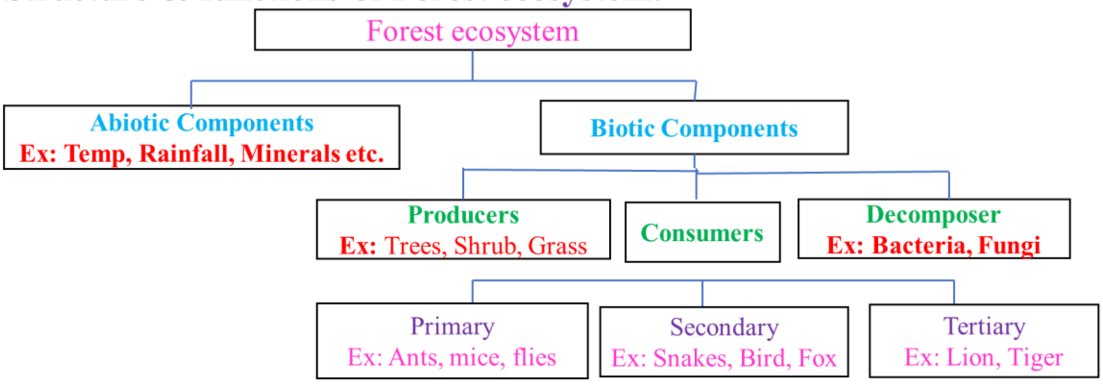
Primary Consumers: – Benthos: These are animals associated with living plants, detrivores and

6	some other microorganisms. **Zooplanktons:** These are chiefly rotifers, protozoans, they feed on phytoplankton. **Secondary Consumers:** They are the Carnivores which feed on herbivores, these are chiefly insect and fish, most insects & water beetles, they feed on zooplanktons. **Tertiary Consumers:** These are some large fish as game fish, turtles, which feed on small fish and thus become tertiary consumers. **Decomposers:** They are also known as microconsumers. They decompose dead organic matter of both producers and animal to simple form. Thus, they play an important role in the return of minerals again to the pond ecosystem, they are chiefly bacteria, & fungi **2.River or Stream ecosystem:** As Compared with lentic freshwater (Ponds & lakes), lotic waters such as streams, and river have been less studied. However, the various components of a riverine and stream ecosystem can be arranged as follows. **Producers**: The chief producers that remain permanently attached to a firm substratum are green algae as Cladophora, and aquatic mosses. **Consumers:** The consumers show certain features as permanent attachment to firm substrata, presence of hooks & suckers, sticky undersurface, streamline bodies, flattened bodies. Thus, a variety of animal are found, which are fresh spongy and caddis-fly larvae, snails, flat worms etc. **Decomposers:** Various bacteria and fungi like actinomycetes are present which acts as decompose		
6	**Explain about Forest ecosystem.**	**Level-2**	**CO1**
Ans.	Forests occupy roughly 40 % of the land. In India these occupy roughly one-tenth of the total land area. The different components of forest ecosystem include: **Abiotic components**: Include inorganic and organic substances present in soil and atmosphere. Dead organic debris is also present in forests. **Biotic components**: These include producers like trees, shrubs and ground vegetation. Among the primary consumers are the herbivores which include animals feeding on tree leaves as ants, flies, beetles, spiders etc. and larger animals like elephants, deer etc. Secondary consumers are the carnivores like snakes, birds and fox etc. Tertiary consumers are the top carnivores like lion, tiger etc. Decomposers include microorganisms like fungi, bacteria and actinomycetes. Characteristics of Forest ➢ A forest is a living community of various species of trees and small form of vegetation characterized by warm temp and adequate rainfall.		

- it regulates the climate and rainfall.
- rich in organic matter and nutrient of forest soils.

There are four types of forests: Deciduous, Tropical, coastal and coniferous forests.

Structure & functions of Forest ecosystem:



PART–C 10 Marks Questions

1	**Explain the deserts ecosystem.**	**Level-3**	**CO1**

Ans. Desert occupy about 17% of land, occurring in the regions with an annual rainfall of about 25 centimetres. The species composition of such an ecosystem is varied and typical due to the extremes of temperature and water factors.

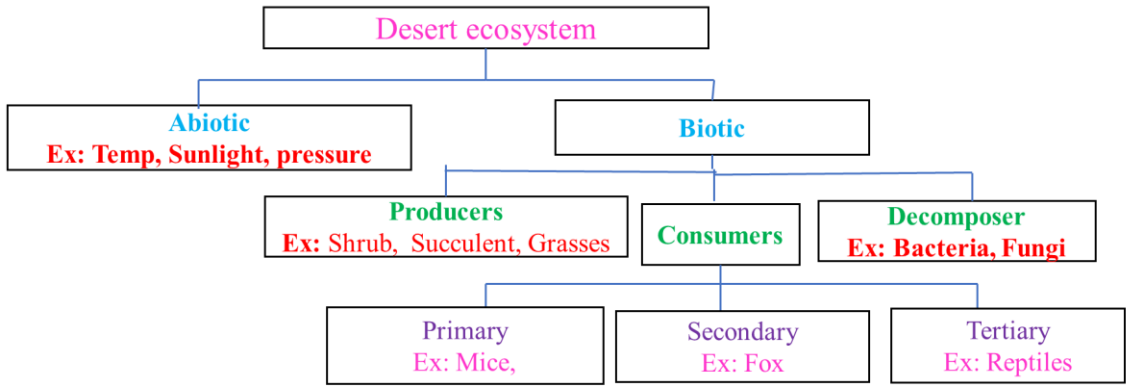
The biotic components include:

- Producers are shrubs, especially bushes, some grasses and a few trees.
- Consumers include animals like reptiles and insects which are able to live under xeric conditions. In addition, there are some nocturnal rodents and birds. The camel feeds on tender shoots of the plants.
- Decomposers are very few, as due to poor vegetation the amount of dead organic matter is correspondingly less. They are some fungi and bacteria.

Characteristics:

- The desert air in dry and the climate is hot.
- Characterized by less than 25% rain fall.
- Vegetation is very less. Because the soil is very poor in nutrients and organic matter.

Structure & functions of Desert ecosystem:



2	**Write about Pond ecosystem.**	**Level-3**	**CO1**

Ans.

A Pond as a whole serves a good example of freshwater ecosystem.

Abiotic Components: The chief components are heat, light, P^H of water, CO_2, oxygen, calcium, nitrogen, phosphates, etc.

Biotic Components: The various organization that constitute the biotic component are as follows,

Producers: These are green plants, and some photosynthetic bacteria. The producer fixes radiant energy and converts it into organic substances as carbohydrates, protein etc.

Macrophytes: these are large rooted plants, which include partly or completely submerged hydrophytes, eg Hydrilla, Trapha, Typha.

P**hytoplankton**: These are minute floating or submerged lower plants eg algae.

Consumers: They are heterotrophs which depend for their nutrition on the organic food manufactured by producers.

Primary Consumers: – Benthos: These are animals associated with living plants, detrivores and some other microorganisms.

Zooplanktons: These are chiefly rotifers, protozoans, they feed on phytoplankton.

Secondary Consumers: They are the Carnivores which feed on herbivores, these are chiefly insect and fish, most insects & water beetles, they feed on zooplanktons.

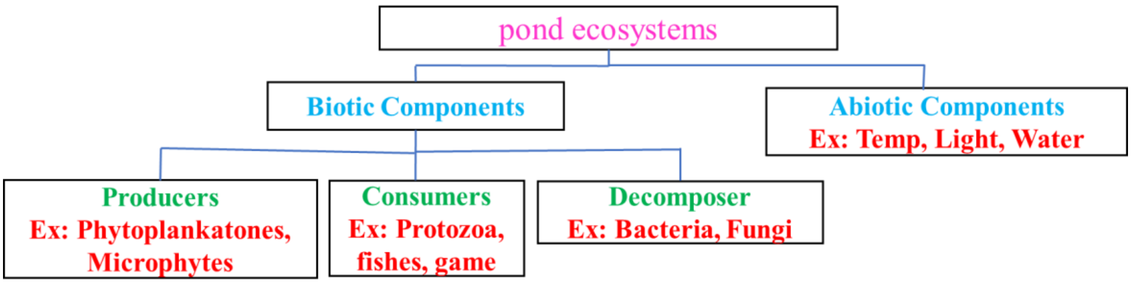
Tertiary Consumers: These are some large fish as game fish, turtles, which feed on small fish and thus become tertiary consumers.

Decomposers: They are also known as microconsumers. They decompose dead organic matter of both producers and animal to simple form. Thus, they play an important role in the return of minerals again to the pond ecosystem, they are chiefly bacteria, & fungi.

Characteristics:

- It is a small fresh water aquatic ecosystem.
- Pond is temporary, only seasonal.
- It is stored or stagnant water body
- Ponds get polluted easily due to limited amount of water.

component & function of pond ecosystems.



3	**Explain the Bio Geo Chemical cycles.**	**Level-2**	**CO1**

Ans. The cyclic flow of nutrients between the biotic and abiotic Component is known as nutrient cycle or bio- geo chemical cycle.

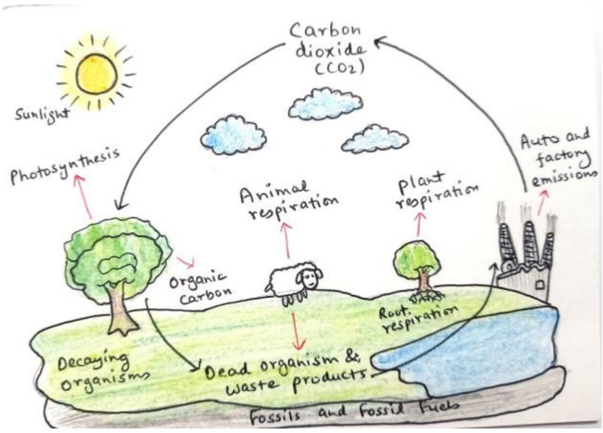
Nutrients: The elements, which are essential for the survival both plant and animals called nutrient. The Major are nutrients like C, H, O and N are cycled again and again between biotic and abiotic components in the ecosystem.

1. The Carbon cycle

The physical cycle of carbon through the Earth's biosphere, geosphere, hydrosphere and atmosphere that includes such processes as photosynthesis, decomposition, respiration and carbonification. The carbon cycle describes the flow of carbon between the biosphere, the geosphere, and the atmosphere, and is essential to maintaining life on earth. Atmospheric Carbon Dioxide: Carbon in the earth's atmosphere exists in two main forms:

carbon dioxide and methane. Carbon dioxide leaves the atmosphere through photosynthesis, thus entering the terrestrial and marine biospheres. Carbon dioxide also dissolves directly from the atmosphere into bodies of water (oceans, lakes, etc.), as well as dissolving in precipitation as raindrops fall through the atmosphere. When dissolved in water, carbon dioxide reacts with water molecules and forms carbonic acid, which contributes to ocean acidity. Human activity over the past two centuries has significantly increased the amount of carbon in the atmosphere, mainly in the form of carbon dioxide, both by modifying ecosystem's ability to extract carbon dioxide from the atmosphere and by emitting it directly, e.g. by burning fossil fuels and manufacturing concrete.

Herbivorous animals feed on plant material, which is used by them for energy and for their growth. Both plants and animals release carbon dioxide during respiration. They also return fixed carbon to the soil in the waste they excrete. When plants and animals die, they return their carbon to the soil. These processes complete the carbon cycle.



4	**Write the scope and importance of ecosystem**	**Level-4**	**CO1**

Ans.

Scope of ecosystems:

- Helps in to tackle problems like pollution, floods, O3 depletion, global warming.
- It is necessary in maintaining ecological balance and understanding different cycles (oxygen, nitrogen, carbon etc.)
- Helps in protecting flora and fauna
- We can maintain balance in nature and can prevent ecological disaster
- Played an Important role in human welfare, agriculture, conservation of wild life.
- Ecology in the scientific study of interaction between organisms and the environment, which intern determine both the distribution of organisms and their abundance.

Importance of ecosystem

- Ecosystem Services are the natural services or earth capital that support life on the earth.
- Essential to the quality of human life and to this functioning of the world economics.
- In the formation of soil, minerals, water bodies.
- For the formation of bio-geo chemical cycles.
- There is no equilibrium in the nature without eco system.

Unit 2 Question bank

Part--- A 2 Marks Questions			
1	**Mention any two effects on humans by using mineral resources**	**Level-2**	**CO2**
Ans.	Exposure to minerals like asbestos can lead to lung diseases such as asbestosis or cancer. Mining can lead to the pollution of nearby water sources, which causes waterborne diseases.		
2	**What are renewable energy resources?**	**Level-1**	**CO2**
Ans.	Renewable energy resources are those that are replenished naturally and are sustainable over time. **Examples:** Solar energy, wind energy, and hydropower.		
3	**What is mineral?**	**Level-1**	**CO2**
Ans.	A mineral is a naturally occurring, inorganic solid with a definite chemical composition and crystalline structure.		
4	**Write about flood.**	**Level-2**	**CO2**
Ans.	A flood is the overflow of water onto normally dry land, usually caused by heavy rainfall, river overflow, or melting snow. It leads to loss of life, property damage, and environmental destruction.		
5	**Why are fossil fuels regarded as non-renewable energy resources?**	**Level-2**	**CO2**
Ans.	Fossil fuels are considered non-renewable because they are formed over millions of years, and the rate at which they are being consumed is much higher than their natural replenishment rate.		
6	**Define mineral. Name the minerals available in India.**	**Level-2**	**CO2**
Ans.	A mineral is a naturally occurring substance with a specific chemical composition. **Examples of minerals in India**: Iron ore, Coal, Bauxite		
7	**What is the environmental impact of mining?**	**Level-4**	**CO2**
Ans.	Mining can lead to environmental damage such as deforestation, habitat destruction, and pollution of water sources from waste disposal.		
8	**Give the alternate energy sources.**	**Level-1**	**CO2**
Ans.	Alternate energy sources include solar energy, wind energy, hydroelectric power, and biomass.		
9	**Distinguish renewable and non-renewable energy resources.**	**Level-1**	**CO2**
Ans.	**Renewable Energy Resources:** These are energy sources that are naturally replenished on a human timescale. Examples include solar, wind, hydro, geothermal, and biomass. They are considered sustainable because they do not deplete over time. **Non-Renewable Energy Resources**: These are energy sources that cannot be replenished within a human timescale. Examples include coal, oil, natural gas, and nuclear energy (uranium). They		

	take millions of years to form and will eventually run out.		
10	**Solar Energy Harvesting Devices**	**Level-2**	**CO2**
Ans.	Photovoltaic (PV) Cells: These devices convert sunlight directly into electricity through the photovoltaic effect. When sunlight hits the semiconductor material (typically silicon), it excites electrons, creating an electric current. Solar Thermal Collectors: These capture sunlight to produce heat, which can then be used for water heating, space heating, or even electricity generation.		
PART–B 5 Marks Questions			
1	**Explain about forest resources**	**Level-1**	**CO2**
Ans.	Forests are a vital natural resource, providing a range of ecological, economic, and social benefits essential for life on Earth. They cover about 31% of the world's land area and play a critical role in supporting biodiversity, regulating the climate, and offering resources to humans. Here's a detailed explanation of their importance: **Biodiversity**: Forests are home to a vast variety of plants, animals, and microorganisms. They support around 80% of the world's terrestrial species, making them essential for maintaining global biodiversity. Different forest types, such as tropical rainforests and boreal forests, host unique ecosystems that sustain diverse flora and fauna. **Climate Regulation**: Forests act as carbon sinks, absorbing significant amounts of carbon dioxide from the atmosphere, which helps mitigate climate change. Through photosynthesis, trees store carbon, which reduces greenhouse gases and helps in temperature regulation globally. Forests also influence rainfall patterns and prevent soil erosion, helping to maintain local climate stability. **Economic Resources**: Forests provide timber, fuelwood, medicinal plants, fruits, and other non-timber products that are vital for both local economies and global trade. Many communities rely on forests for their livelihoods, with timber being a major industry in numerous countries. Forests also support eco-tourism, offering recreational and educational experiences for people worldwide. **Environmental Benefits**: Forests help protect water resources by filtering and regulating water flow in rivers and streams. They reduce the risk of natural disasters like floods and landslides by stabilizing the soil. Forests also improve air quality by filtering pollutants and providing oxygen, which is essential for human and animal life. **Social and Cultural Importance**: Many indigenous communities have lived in harmony with forests for centuries, with cultural and spiritual practices deeply connected to forested lands.		

	Forests provide traditional medicines and materials for these communities and are part of their identity. Conservation of forests is crucial for preserving indigenous heritage and knowledge. However, forests are increasingly threatened by deforestation, illegal logging, and land conversion for agriculture and urban development. Sustainable forest management practices are needed to protect and restore forest resources, ensuring their benefits for future generations.		
2	**Write the merits and demerits associated with a) solar energy b) Wind energy**	**Level-2**	**CO2**
Ans.	**a) Solar Energy** **Merits:** 1. **Renewable and Abundant:** Solar energy is an inexhaustible source, widely available and sustainable. 2. **Environmentally Friendly:** It produces no pollution or greenhouse gases, helping reduce the carbon footprint. 3. **Low Operating Costs:** After installation, solar panels require minimal maintenance and have low operational costs. 4. **Energy Independence:** Solar energy can reduce dependency on fossil fuels, enhancing energy security for users. 5. **Versatile Applications:** Solar energy can be used for electricity, heating, and even powering remote or off-grid areas. **Demerits:** 1. **Weather-Dependent:** Solar energy production drops significantly on cloudy or rainy days and is unavailable at night. 2. **High Initial Costs:** The cost of solar panels, inverters, and installation can be a financial barrier. 3. **Space Requirements:** Solar panels require significant space to produce enough energy, which may not be feasible for all areas. 4. **Energy Storage Costs:** Effective storage solutions, like batteries, are expensive and may limit efficient energy use. 5. **Manufacturing Impact:** The production of solar panels involves chemicals and processes that may be harmful to the environment. **b) Wind Energy** **Merits:**		

	1. **Renewable and Sustainable:** Wind is an inexhaustible resource and can be harnessed consistently in windy areas. 2. **Eco-Friendly:** Wind turbines produce no emissions during operation, making them a clean energy source. 3. **Low Operational Costs:** Once installed, wind turbines have minimal operational and maintenance costs. 4. **Job Creation:** Wind energy projects create jobs in manufacturing, installation, and maintenance. 5. **Land Use Efficiency:** Wind turbines occupy only a small portion of land, allowing the remaining space to be used for farming or other activities. **Demerits:** 1. **Intermittent Power Supply:** Wind energy production depends on wind speeds, making it less reliable and predictable. 2. **Noise Pollution:** Wind turbines generate noise, which can be disruptive to nearby residents. 3. **Threat to Wildlife:** Wind farms can harm birds and bats, especially in migration pathways. 4. **High Initial Investment:** The cost of wind turbine technology and installation can be high. 5. **Aesthetic Impact:** Large wind farms may alter natural landscapes, affecting the visual appeal of areas. Both solar and wind energy are essential for a sustainable future, but challenges must be addressed to maximize their benefits and reduce their limitations.		
3	**What are the fresh water resources?**	**Level-1**	**CO2**
Ans.	Fresh water resources are essential for life, supporting ecosystems, agriculture, drinking water supplies, and industrial processes. They include all sources of water with low salt concentrations (less than 1%), suitable for human consumption and other vital uses. The main types of fresh water resources are: 1. **Rivers and Streams**: These are surface water resources that flow from higher to lower elevations. They play a crucial role in providing water for drinking, irrigation, transportation, and recreation. Major rivers also contribute to the water supply for large populations and support diverse ecosystems.		

4	2. **Lakes and Ponds**: These are natural or man-made water bodies that store fresh water. Lakes and ponds help maintain local biodiversity, support agriculture, and provide water for industrial and domestic use. They also play a significant role in groundwater recharge. 3. **Glaciers and Ice Caps**: Glaciers and polar ice caps store about 68% of the world's fresh water. Though frozen, they release water seasonally through melting, feeding rivers and lakes. Glacier melt is an essential source of water for regions near mountainous areas, especially during warmer months. 4. **Groundwater**: Groundwater is found beneath the Earth's surface in soil and rock layers known as aquifers. It represents about 30% of the world's fresh water and is a primary source for drinking and irrigation. Groundwater is accessed through wells and boreholes, and its availability is crucial for regions with limited surface water. 5. **Rainwater**: Rain is a direct source of fresh water, replenishing rivers, lakes, and groundwater. Rainwater harvesting, the collection and storage of rain, is a technique used to increase water availability in areas with scarce water resources. 6. **Wetlands**: Wetlands, such as marshes, swamps, and bogs, store fresh water and are rich in biodiversity. They help in natural water purification, flood control, and groundwater replenishment, making them essential for both human and ecological health. These fresh water resources are limited and unevenly distributed, making sustainable management crucial to ensure

4	**Briefly explain the control of droughts and floods?**	**Level-2**	**CO2**
Ans.	Managing droughts and floods involves proactive strategies to reduce the impact of these natural events. Here's a brief overview: **Control of Droughts** 1. **Water Conservation**: Encouraging efficient water use and reducing wastage through practices like rainwater harvesting, recycling wastewater, and promoting water-saving technologies helps conserve water in times of scarcity. 2. **Improved Irrigation Techniques**: Methods such as drip irrigation and sprinkler systems use less water compared to traditional methods, preserving water in agriculture. 3. **Afforestation and Soil Conservation**: Planting trees and maintaining soil health helps retain water in the soil and reduces evaporation, making more water available during dry periods. 4. **Groundwater Recharge**: Techniques like building check dams, percolation tanks, and		

	recharge wells help increase groundwater levels, creating a reserve that can be used during droughts. 5. **Drought Monitoring and Early Warning Systems**: Monitoring rainfall, soil moisture, and river flows helps anticipate drought conditions, allowing authorities to take timely measures to minimize impacts. **Control of Floods** 1. **Dams and Reservoirs**: Dams store excess water from rivers and can regulate water release, preventing downstream flooding. Reservoirs can hold floodwater and release it gradually. 2. **River Channelization**: Modifying rivers by deepening, widening, or redirecting channels can help manage water flow, reducing the risk of overflow during heavy rains. 3. **Construction of Embankments and Levees**: Building barriers along riverbanks helps contain rising water levels, preventing water from spilling into adjacent lands. 4. **Drainage Systems and Floodway's**: Creating efficient drainage networks and designated floodway's diverts floodwaters away from populated areas, reducing flood damage. 5. **Floodplain Management**: Limiting development in flood-prone areas and restoring wetlands helps control floodwaters naturally, as wetlands absorb excess water and slow its movement. Both drought and flood control rely on long-term planning, infrastructure, and sustainable practices to reduce vulnerability and ensure water security.		
5	**Distinguish renewable and non-renewable energy resources?**	**Level-1**	**CO2**
Ans.	Renewable and non-renewable energy resources are distinguished primarily by their availability and the rate at which they can be replenished. Here's a breakdown: **Renewable Energy Resources** 1. **Definition**: Renewable energy resources are those that are naturally replenished on a human timescale and can be used repeatedly without depletion. 2. **Examples**: Solar energy, wind energy, hydropower, geothermal energy, and biomass. 3. **Availability**: These resources are abundant and can be sustainably harvested without running out. 4. **Environmental Impact**: Renewable energy sources produce little to no greenhouse gases during use, making them eco-friendly and essential in reducing carbon emissions. 5. **Limitations**: Some renewable sources, like solar and wind, depend on weather		

	conditions, which can make them intermittent. Storage solutions, like batteries, may be needed for consistent supply. 6. **Sustainability**: Since these sources regenerate naturally, they are sustainable long-term options for energy production. **Non-Renewable Energy Resources** 1. **Definition**: Non-renewable energy resources are finite and cannot be replenished on a human timescale. Once used, they are depleted permanently. 2. **Examples**: Coal, oil, natural gas, and uranium (used in nuclear energy). 3. **Availability**: These resources are limited and will eventually run out if extracted at a high rate. 4. **Environmental Impact**: Burning fossil fuels releases high levels of greenhouse gases and pollutants, contributing significantly to air pollution and climate change. Nuclear energy produces radioactive waste, which requires careful disposal. 5. **Limitations**: Non-renewable resources are extracted from the Earth and are increasingly difficult and costly to obtain as reserves deplete. 6. **Sustainability**: These resources are not sustainable long-term due to their finite nature and environmental impact. Renewable energy is sustainable, eco-friendly, and naturally replenishing, whereas non-renewable energy is finite, environmentally harmful, and faces eventual depletion. Transitioning to renewable sources is key to a sustainable energy future.		
6	**Define desertification and write about its effects?**	**Level-2**	**CO2**
Ans.	Desertification is the process by which fertile land becomes increasingly arid and unproductive, often turning into desert-like conditions. This phenomenon occurs primarily due to human activities, such as deforestation, unsustainable farming practices, overgrazing, and climate change. Desertification severely impacts ecosystems, economies, and human livelihoods, especially in arid and semi-arid regions. **Effects of Desertification** 1. **Loss of Arable Land**: Productive soil becomes barren, reducing agricultural output. This can lead to food insecurity, as less land is available for growing crops and supporting livestock. 2. **Biodiversity Loss**: Desertification leads to the destruction of habitats, threatening plant and animal species that rely on healthy land. Reduced biodiversity affects ecosystem resilience, making it harder for ecosystems to recover.		

	3. **Increased Poverty and Food Scarcity**: Communities that depend on agriculture for their livelihood suffer economically due to reduced crop yields and livestock productivity. This can lead to poverty, malnutrition, and migration from affected areas. 4. **Climate Change Acceleration**: Degraded land stores less carbon and often releases previously stored carbon back into the atmosphere, contributing to greenhouse gas emissions. Vegetation loss also reduces the land's ability to absorb carbon dioxide, worsening climate change. 5. **Water Scarcity and Decreased Water Quality**: Desertification lowers soil moisture levels, reducing groundwater recharge and water availability. It also increases water salinity and contamination, affecting water quality for drinking, irrigation, and sanitation. 6. **Soil Erosion and Dust Storms**: As vegetation is lost, the soil becomes more susceptible to erosion by wind and water. This leads to dust storms, which can damage crops, pollute air quality, and contribute to respiratory problems in humans and animals. Desertification poses serious ecological, economic, and social challenges. Addressing it requires sustainable land management practices, reforestation, and measures to combat climate change to preserve soil health and protect vulnerable regions from degradation.
PART–C 10 Marks Questions	

1	**What are the benefits and problems of construction of dams?**	**Level-3**	**CO2**
Ans.	Dams are large structures built across rivers or streams to store and control water flow. They provide numerous advantages but also come with significant drawbacks. Here’s an outline of the main benefits and problems associated with dam construction: **Benefits of Dams** 1. **Water Storage for Domestic and Industrial Use**: Dams store large quantities of water, providing a reliable water supply for drinking, household use, and industrial processes, especially in arid and semi-arid regions. 2. **Irrigation Support**: Stored water can be released to irrigate agricultural land, increasing crop yields and supporting food production. Dams have played a critical role in agricultural growth and food security worldwide. 3. **Hydroelectric Power Generation**: Many dams are equipped with turbines to generate hydroelectric power, a clean and renewable energy source that reduces dependency on fossil fuels and helps decrease greenhouse gas emissions.		

	4. **Flood Control**: Dams regulate river flow, helping prevent or reduce the impact of floods during heavy rainfall by holding back excess water. This protects downstream areas and reduces flood damage to communities and infrastructure. 5. **Recreation and Tourism**: Dams create reservoirs that support recreational activities like boating, fishing, and tourism. These activities can boost local economies and offer leisure opportunities for residents and visitors. **Problems of Dams** 1. **Displacement of Communities**: Building dams often requires the flooding of large areas, which displaces local communities and disrupts livelihoods. People may lose homes, farms, and access to resources, leading to social and economic challenges. 2. **Environmental Impact**: Dams alter natural river ecosystems, affecting fish migration, disrupting aquatic habitats, and reducing biodiversity. The change in water flow and sediment distribution can also impact the surrounding wildlife and vegetation. 3. **Risk of Reservoir-Induced Seismicity**: The weight of large reservoirs and the change in water pressure can sometimes trigger earthquakes, especially in geologically active areas. This poses a potential risk to nearby regions. 4. **Sedimentation**: Sediment that would naturally flow downstream is trapped in reservoirs, which reduces dam storage capacity over time. This can also cause erosion downstream and reduce soil fertility in deltas, affecting agricultural productivity. 5. **Water Quality and Health Concerns**: Stagnant water in reservoirs can become a breeding ground for waterborne diseases and invasive species. Reduced water quality can also affect drinking water supplies and harm aquatic life. 6. **High Costs and Long-Term Maintenance**: Dam construction and maintenance require significant investment and resources. The initial cost is high, and continuous maintenance and monitoring are essential to ensure structural safety and functionality. While dams offer benefits like water storage, flood control, and renewable energy, they also have complex social, environmental, and economic impacts. Sustainable planning and careful consideration of affected ecosystems and communities are essential for balancing these benefits and challenges.
2	**What are the problems associated with over utilization of ground water?** **Level-2** **CO2**
Ans.	Groundwater is a crucial resource for drinking water, agriculture, and industrial use. However, excessive groundwater extraction can lead to a range of environmental, social, and economic

problems. Here's an outline of the main issues caused by over-utilization of groundwater:

1. Depletion of Water Tables

- Excessive extraction of groundwater lowers the water table, making it difficult and costly to access water. In some regions, wells have to be dug deeper to reach diminishing water levels, which can be prohibitively expensive.

2. Land Subsidence

- When groundwater is removed faster than it can be naturally replenished, the land above may sink or collapse in a process known as subsidence. This is particularly common in urban and agricultural areas with high water demand, damaging infrastructure, roads, and buildings.

3. Reduced Water Quality

- Over-utilization can lead to the intrusion of contaminants, such as salts and chemicals, especially in coastal areas where saltwater may enter freshwater aquifers (known as saltwater intrusion). Polluted groundwater requires costly treatment to make it usable again.

4. Loss of Surface Water Connections

- Groundwater is often connected to surface water bodies like rivers, lakes, and wetlands. Excessive extraction can reduce the flow of these surface water resources, harming ecosystems that depend on them and reducing water availability for other users.

5. Decline in Agricultural Productivity

- Many agricultural regions rely heavily on groundwater for irrigation. When water levels drop, it becomes more challenging and expensive to irrigate crops, which can lead to reduced agricultural yields and impact food security.

6. Ecological Impact on Wetlands and Wildlife

- Wetlands and some types of forests rely on stable groundwater levels. Overuse of groundwater can dry up these habitats, endangering plant and animal species that depend on them. The loss of wetlands also reduces biodiversity and weakens ecosystem resilience.

7. Increased Energy Costs

- As groundwater levels fall, more energy is required to pump water from greater depths. This increases the cost of water extraction for farmers, industries, and households, raising overall energy consumption and expenses.

8. Long-Term Irreversibility

	• Overdrawn aquifers, especially those in arid and semi-arid regions, may take centuries to naturally recharge, if at all. This makes groundwater a finite resource in many regions, limiting its availability for future generations.		
3	**Explain briefly about water resources?**	**Level-1**	**CO2**
Ans.	Land resources encompass the various natural resources found on the Earth's surface, which are crucial for supporting life, agriculture, and human development. They include soil, forests, minerals, and land itself, and their sustainable management is essential for environmental health and economic stability. Here are some key aspects of land resources: **1. Types of Land Resources** • **Soil**: The upper layer of earth that supports plant life, soil is vital for agriculture. It provides nutrients and water to crops and plays a crucial role in the ecosystem, supporting biodiversity and influencing water cycles. • **Forests**: Forests are significant land resources that provide timber, fuelwood, and non-timber forest products like fruits, nuts, and medicinal plants. They also play a critical role in carbon sequestration, biodiversity conservation, and watershed protection. • **Minerals**: Land resources include various mineral deposits, such as metals, coal, and industrial minerals, which are essential for construction, energy production, and manufacturing. • **Pasture Land**: Areas of land used for grazing livestock are vital for food production. Sustainable management of pasture land helps maintain soil health and supports biodiversity. **2. Importance of Land Resources** • **Agriculture**: Land is a fundamental resource for food production, providing the space and nutrients needed to grow crops and raise animals. Sustainable land use practices are crucial for food security. • **Economic Development**: Land resources support various economic activities, including agriculture, forestry, mining, and tourism. Proper management can drive economic growth and job creation. • **Ecosystem Services**: Land resources provide essential ecosystem services, such as water filtration, carbon storage, and habitat for wildlife. Healthy land ecosystems are vital for maintaining environmental balance. **3. Challenges Facing Land Resources** • **Land Degradation**: Practices like deforestation, overgrazing, and unsustainable		

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agriculture can lead to soil erosion, loss of fertility, and desertification, reducing the land's productivity.
Urbanization: Rapid urban development often leads to the conversion of agricultural and natural land into built environments, threatening ecosystems and reducing available land for food production.
Pollution: Contamination from agricultural chemicals, industrial waste, and urban runoff can degrade land quality and pose risks to human health and the environment.

4. Sustainable Management Practices

Conservation Agriculture: Techniques such as crop rotation, cover cropping, and reduced tillage help maintain soil health and enhance productivity while minimizing environmental impact.
Reforestation and Afforestation: Planting trees in deforested areas or creating new forests helps restore ecosystems, sequester carbon, and improve biodiversity.
Land Use Planning: Effective land use planning considers the balance between development, conservation, and sustainable resource use to ensure long-term land health and availability.

Land resources are fundamental for sustaining human life and economic activity. Their responsible and sustainable management is crucial for ensuring food security, supporting ecosystems, and mitigating the impacts of climate change. Addressing challenges such as land degradation, pollution, and urbanization is vital for preserving land resources for future generations.
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<tr><td>4</td><td>What is the environmental impact of mining?</td><td>Level-3</td><td>CO2</td></tr>
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Mining activities can have significant and lasting effects on the environment, impacting ecosystems, air and water quality, and local communities. The extraction of minerals, metals, and fossil fuels often involves large-scale operations that can lead to various environmental issues. Here’s an overview of the major environmental impacts associated with mining:

1. Land Degradation

Deforestation: Mining operations often require clearing vast areas of land, leading to deforestation and habitat destruction. This can disrupt local ecosystems and lead to the loss of biodiversity.
Soil Erosion: The removal of vegetation exposes soil to erosion by wind and water. Erosion can lead to sedimentation in rivers and lakes, degrading water quality and affecting aquatic life.

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2. Water Pollution

- **Contaminated Water Sources**: Mining activities can release harmful chemicals, such as heavy metals (lead, mercury, arsenic) and acids, into nearby water bodies. This contamination can affect drinking water supplies, harm aquatic ecosystems, and pose health risks to local communities.
- **Acid Mine Drainage**: When sulfide minerals are exposed to air and water during mining, they can produce sulfuric acid, leading to acid mine drainage. This can severely impact water quality in surrounding areas, harming aquatic life and making water unsafe for consumption.

3. Air Pollution

- **Dust Emissions**: Mining activities generate significant dust, which can degrade air quality and pose health risks to nearby populations. Dust can contain harmful particles and metals that may affect respiratory health.
- **Fugitive Emissions**: The combustion of fossil fuels in mining operations can lead to the release of greenhouse gases and other pollutants into the atmosphere, contributing to climate change and local air quality issues.

4. Habitat Destruction

- **Displacement of Wildlife**: Mining disrupts habitats, leading to the displacement of flora and fauna. Many species may struggle to survive if their natural habitats are altered or destroyed.
- **Fragmentation of Ecosystems**: Mining can create physical barriers that fragment ecosystems, making it difficult for species to migrate, reproduce, and maintain genetic diversity.

5. Resource Depletion

- **Loss of Biodiversity**: Mining can lead to the extinction of plant and animal species, particularly in biodiversity-rich areas. The destruction of habitats can permanently alter ecosystems and reduce their resilience to environmental changes.

6. Socioeconomic Impacts

- **Displacement of Communities**: Mining often leads to the displacement of local communities, disrupting livelihoods and cultural practices. This can create social tensions and lead to loss of traditional knowledge and practices.
- **Economic Inequality**: While mining can create jobs and contribute to local economies, it may also lead to economic disparities. Wealth generated from mining often does not

	benefit local communities, exacerbating poverty and inequality. **7. Long-term Environmental Consequences** • **Land Rehabilitation Challenges**: After mining operations cease, restoring the land to its original state can be difficult and expensive. Many areas remain degraded, impacting future land use and ecosystem recovery. • **Ongoing Pollution**: The effects of mining can persist long after operations have ended, as pollutants can continue to leach into the environment and disrupt ecosystems and human health for decades. The environmental impact of mining is profound and multifaceted, affecting land, water, air, and local communities. To mitigate these impacts, sustainable mining practices and regulations are essential. These may include responsible resource extraction, rehabilitation of mined areas, monitoring of environmental effects, and engaging local communities in decision-making processes. Sustainable practices can help balance economic benefits with environmental protection, ensuring that mining contributes to long-term ecological and community health.

Unit 3 Question bank

Part--- A 2 Marks Questions			
1	**What is biodiversity?**	**Level-1**	**CO3**
Ans.	Biodiversity refers to the variety of life forms, ecosystems, and genetic diversity on Earth.		
2	**What are the main threats to biodiversity?**	**Level-1**	**CO3**
Ans.	Habitat destruction, climate change, pollution, overexploitation, and invasive species are the main threats.		
3	**What is an endangered species?**	**Level-1**	**CO3**
Ans.	An endangered species is a species at risk of extinction due to various factors, such as habitat loss or poaching.		
4	**What is the role of keystone species?**	**Level-1**	**CO3**
Ans.	Keystone species play a critical role in maintaining the structure and function of an ecosystem.		
5	**What is a biodiversity hotspot?**	**Level-1**	**CO3**
Ans.	A biodiversity hotspot is a region with a high level of endemic species that is under threat from human activity.		
6	**How does climate change affect biodiversity?**	**Level-1**	**CO3**
Ans.	Climate change alters habitats, disrupts species migration patterns, and can lead to the extinction of vulnerable species.		
7	**What is sustainable development?**	**Level-1**	**CO3**
Ans.	Sustainable development is growth that meets current needs without compromising the ability of future generations to meet theirs, including maintaining biodiversity.		
8	**What is In-Situ conservation? What is Ex-Situ conservation?**	**Level-1**	**CO3**
Ans.	In-Situ conservation is the conservation of species in their natural habitats, such as national parks or wildlife reserves. Ex-Situ conservation is the conservation of species outside their natural habitats, such as in botanical gardens or zoos.		
9	**What is the National Biodiversity Act?**	**Level-1**	**CO3**
Ans.	The National Biodiversity Act of India provides a framework for the conservation, sustainable use, and equitable sharing of biological resources.		
10	**What is poaching?**	**Level-1**	**CO3**

Ans.	Poaching is the illegal hunting, capturing, or killing of wildlife, often for profit.		
PART–B 5 Marks Questions			
1	What is biodiversity? Explain its types.	Level-2	CO3
Ans.	Biodiversity refers to the variety of life forms present on Earth, including plants, animals, microorganisms, their genes, and the ecosystems they form. It can be categorized into three main types 1. **Genetic Diversity:** This is the variation in genetic material within species. It allows species to adapt to environmental changes and survive in the long term. For example, genetic diversity in crops ensures resistance to diseases and pests. 2. **Species Diversity:** This refers to the number and variety of species in an area. Rich species diversity is an indicator of ecosystem health, such as the variety of fish species in coral reefs. 3. **Ecosystem Diversity:** This is the variety of ecosystems on Earth, including forests, grasslands, deserts, oceans, and wetlands. Different ecosystems provide vital services like clean water, carbon storage, and habitat for species. Biodiversity is essential because it supports ecosystem stability, food security, and human well-being.		
2	**Explain the consumptive and productive use values of biodiversity with examples.**	**Level-2**	**CO3**
Ans.	Biodiversity provides several types of benefits that humans rely on. Two important values are: 1. **Consumptive Use Value:** This refers to the direct use of biodiversity for goods that are consumed locally. Examples include food (fruits, vegetables, and fish), fuelwood, and medicinal plants. Indigenous communities often rely on local biodiversity for subsistence. 2. **Productive Use Value:** This includes resources used in the production of goods and services, often for commercial purposes. For instance, timber, rubber from trees, cotton for textiles, and raw materials like oil and biofuels are sourced from biodiversity. Pharmaceuticals also depend on plant and animal compounds for the development of medicines. These values underline the economic importance of biodiversity, demonstrating how it contributes to livelihoods, industries, and sustainable development.		

3	**What are the social and ethical values of biodiversity?**	**Level-3**	**CO3**
Ans.	Biodiversity offers social and ethical values that go beyond material or economic benefits: 1**. Social Value:** Biodiversity plays a key role in the cultural and social practices of communities. It is integral to traditions, beliefs, and social customs. Sacred groves, for example, are areas of natural significance in various cultures. Biodiversity also contributes to social well-being by providing opportunities for ecotourism and outdoor activities. 2**. Ethical Value:** Ethical values refer to the belief that species and ecosystems have an intrinsic right to exist, regardless of their utility to humans. It suggests that humans have a moral obligation to protect biodiversity. The protection of endangered species, such as tigers or pandas, is based on ethical considerations of preserving life forms for their own sake, not just for human use. Both these values promote a deeper connection to nature and advocate for the preservation of biodiversity for future generations.		
4	**What are biodiversity hotspots? Why are they important for conservation?**	**Level-3**	**CO3**
Ans.	Biodiversity hotspots are regions that are both rich in endemic species (species found nowhere else) and are under significant threat from human activities. A region is considered a hotspot if it contains at least 1,500 species of vascular plants as endemics and has lost at least 70% of its original habitat. Examples include the Amazon Rainforest, the Western Ghats in India, and the Philippine Islands. These regions are important for conservation because: 1. They contain a disproportionate amount of Earth's biodiversity, including many species that could become extinct if not protected. 2. Hotspots support unique ecosystems that provide critical services like climate regulation, water purification, and soil fertility. 3. Focusing conservation efforts on hotspots helps maximize the protection of biodiversity with limited resources, making these areas a priority for global conservation efforts. Protecting hotspots is key to ensuring the survival of many species and maintaining ecosystem health.		

5	**Discuss the major threats to biodiversity.**	**Level-4**	**CO3**
Ans.	Biodiversity faces multiple threats, many of which are driven by human activities: 1. **Habitat Loss:** The destruction of natural habitats due to deforestation, urbanization, agriculture, and industrialization leads to the displacement or extinction of species. The clearing of rainforests in the Amazon is a prime example. 2**. Poaching and Illegal Wildlife Trade:** Poaching for animal products, such as ivory, fur, and rhino horns, severely affects wildlife populations. Species like tigers, rhinos, and elephants are at risk of extinction due to illegal hunting. 3. **Man-Wildlife Conflicts:** As human populations expand, wildlife is forced into closer contact with humans. Conflicts arise when animals, such as elephants or tigers, damage crops or pose a threat to human life, leading to retaliatory killings. 4. **Climate Change:** Changes in climate patterns disrupt ecosystems and species habitats, forcing species to migrate or adapt. Species unable to cope with these changes face extinction. 5**. Pollution:** Pollution from plastic, chemicals, and industrial waste contaminates ecosystems, harming wildlife and ecosystems. Marine species, such as sea turtles, are affected by plastic pollution in the oceans. These threats collectively contribute to the ongoing loss of biodiversity, which jeopardizes the health of ecosystems and the services they provide.		
6	**What are the key provisions of the National Biodiversity Act of India?**	**Level-2**	**CO3**
Ans.	The National Biodiversity Act (2002) is an important legal framework aimed at the conservation of biodiversity, sustainable use of biological resources, and equitable sharing of benefits derived from them. Key provisions of the Act include: 1. **Biodiversity Management Committees (BMCs):** The Act mandates the formation of BMCs at the local level to ensure the conservation and sustainable use of biodiversity at the grassroots. 2. **National Biodiversity Authority (NBA):** The NBA is responsible for regulating the access to biological resources and traditional knowledge, ensuring fair benefit-sharing.		

	3. **Conservation of Biodiversity:** The Act emphasizes the creation of protected areas for the conservation of ecosystems, species, and genetic diversity. 4. **Regulation of Bioprospecting:** The Act regulates the access to biological resources for bioprospecting, ensuring that traditional knowledge is protected, and benefits are shared with indigenous communities. The National Biodiversity Act is a significant step toward ensuring the sustainable use and conservation of India's rich biodiversity.
PART–C 10 Marks Questions	

1	**Explain In-Situ conservation and Ex-Situ conservation and provide examples.**	**Level-3**	**CO3**
Ans.	In-Situ conservation refers to the conservation of species in their natural habitats, where they can thrive and evolve naturally. This method is considered the most effective way to protect biodiversity because it maintains the species' ecological interactions and genetic diversity. **Key examples of In-Situ conservation include:** 1**. Protected Areas:** National parks, wildlife sanctuaries, and biosphere reserves are examples of protected areas where biodiversity is conserved in its natural setting. For example, the Yellowstone National Park in the U.S. and the Kaziranga National Park in India. 2**. Wildlife Corridors:** These are areas of habitat that connect fragmented ecosystems, allowing species to move and migrate safely between areas, thus enhancing genetic diversity and preventing inbreeding. In-Situ conservation is crucial because it supports the preservation of natural ecosystems and allows species to maintain their evolutionary processes. Ex-Situ conservation refers to the conservation of species outside their natural habitats, typically in controlled environments. This method is used when species are at risk of extinction in the wild and needs direct intervention. **Examples of Ex-Situ conservation include:**		

	1. Zoos and Aquariums: These institutions house endangered species in controlled environments, where breeding programs can help increase populations. The breeding of the giant panda in zoos is a successful Ex-Situ conservation example. **2. Botanical Gardens:** These gardens preserve endangered plant species in controlled settings. The Royal Botanic Gardens in Kew, U.K., store seeds and plant specimens for research and conservation purposes. **3. Seed Banks:** Seed banks like the Svalbard Global Seed Vault store seeds from endangered plant species for future restoration. Ex-Situ conservation acts as a safeguard, ensuring species are preserved and can be reintroduced into the wild when conditions improve.		
2	**Explain the importance of pollination and its impact on biodiversity.**	**Level-3**	**CO3**
Ans.	Pollination is a critical ecological process that facilitates the reproduction of many plants, including important agricultural crops. Pollinators, such as bees, butterflies, birds, and bats, transfer pollen from one flower to another, aiding in fertilization. Importance of Pollination: 1**. Food Production:** Pollination is essential for the production of many fruits, vegetables, and nuts, which are key to human diets and agricultural economies. Crops like apples, almonds, and coffee depend on pollinators. **2. Biodiversity Maintenance:** Pollination helps maintain biodiversity by supporting the growth of plants that provide food and habitat for many other species. The decline in pollinators can lead to a loss of these plant species, causing cascading effects throughout the food web. 3. Economic Value: Pollination supports the global economy by ensuring the productivity of crops. It has been estimated that pollinators contribute billions of dollars annually to global agricultural production.		
3	**What are the benefits of protecting endangered species?**	**Level-4**	**CO3**
Ans.	Protecting endangered species provides numerous ecological, economic, and ethical benefits: **1. Ecological Balance:** Endangered species often play key roles in their ecosystems. For example, predators like wolves control herbivore populations, preventing overgrazing and maintaining plant diversity.		

	2. Biodiversity Conservation: Endangered species are often indicators of ecosystem health. Their protection ensures the preservation of the ecosystems they inhabit, which in turn benefits other species, including humans. **3. Economic Value:** Many species contribute to ecotourism, generating income and employment for local communities. Iconic species like the Bengal tiger and African elephants attract tourists to national parks. **4. Medicinal and Research Value:** Some endangered species may contain valuable compounds that could lead to medical breakthroughs, such as cancer treatments. 5. Ethical Responsibility: Protecting endangered species reflects a moral obligation to preserve life on Earth for future generations.		
4	**Explain the role of the public in biodiversity conservation.**	**Level-3**	**CO3**
Ans.	Public participation is crucial for successful biodiversity conservation. The role of the public can include: **1. Awareness and Education:** Public awareness campaigns can educate people about the importance of biodiversity, leading to more sustainable lifestyles and responsible consumption patterns. **2. Citizen Science:** Citizens can participate in biodiversity monitoring and data collection, contributing valuable information about species distributions and ecosystem health. **3. Sustainable Practices:** People can engage in sustainable practices such as reducing waste, conserving water, and supporting eco-friendly businesses. Community-led initiatives like tree planting and habitat restoration also play a significant role. **4. Advocacy and Policy Influence:** Public pressure can push governments and organizations to adopt stronger policies for biodiversity conservation, such as lobbying for the creation of protected areas or the regulation of harmful industrial practices.		

Unit 4 Question bank

Part--- A 2 Marks Questions			
1	**Define pollution as per Odum.**	**Level-1**	**CO4**
Ans.	According to Odum (1971), pollution is defined as “an undesirable change in the characteristics of air, water, and land that adversely affects life and creates health hazards for all living organisms on Earth.”		
2	**Name two natural sources of air pollution.**	**Level-1**	**CO4**
Ans.	Two natural sources of air pollution are volcanic eruptions and forest fires.		
3	**Give one example each of a primary and a secondary air pollutant.**	**Level-2**	**CO4**
Ans.	An example of a primary air pollutant is carbon monoxide (CO), and an example of a secondary air pollutant is ozone (O_3).		
4	**What is the chemical formula of sulphur dioxide?**	**Level-1**	**CO4**
Ans.	The chemical formula of sulphur dioxide is SO_2.		
5	**What is eutrophication?**	**Level-2**	**CO4**
Ans.	Eutrophication is the enrichment of water bodies with excess nutrients, mainly nitrogen and phosphorus, leading to dense algal blooms that cause oxygen depletion, fish kills, and loss of biodiversity.		
6	**Mention one disease caused by mercury in water.**	**Level-1**	**CO4**
Ans.	One disease caused by mercury in water is Minamata Disease.		
7	**What is the normal range of dissolved oxygen (DO) in water?**	**Level-1**	**CO4**
Ans.	The normal range of dissolved oxygen (DO) in water is 5–8 mg/L		
8	**What is the threshold of pain for noise level in dB?**	**Level-1**	**CO4**
Ans.	The threshold of pain for noise level is approximately 120 dB.		
9	**Name one source of marine oil pollution.**	**Level-1**	**CO4**
Ans.	One source of marine oil pollution is oil spills during transportation or storage.		
10	**What does the 3Rs stand for in waste management?**	**Level-1**	**CO4**
Ans.	The 3Rs in waste management stand for Reduce, Reuse, and Recycle.		
PART–B 5 Marks Questions			
1	**Differentiate between natural and anthropogenic pollution with examples.**	**Level-2**	**CO4**
Ans.	Natural pollution occurs due to natural events without human intervention. It includes pollutants released through processes such as volcanic eruptions, forest fires, soil erosion, cosmic radiation,		

	and ultraviolet (UV) radiation. These are part of Earth's natural cycles. Example: Volcanic eruptions release ash and gases like sulphur dioxide into the atmosphere, causing air pollution. Anthropogenic pollution, on the other hand, is caused by human activities such as industrialization, urbanization, transportation, and improper waste disposal. This type of pollution is increasing with development and poses serious threats to the environment and health. Example: Burning fossil fuels in factories and vehicles emits carbon monoxide (CO) and nitrogen oxides (NO_x), contributing to air pollution and smog formation.		
2	**Write a short note on acid rain and its formation.**	**Level-2**	**CO4**
Ans.	Acid rain is a type of secondary air pollution formed when sulphur dioxide (SO_2) and nitrogen oxides (NO_x) released from burning fossil fuels react with water vapour in the atmosphere. These reactions produce sulphuric acid (H_2SO_4) and nitric acid (HNO_3), which fall to the ground with rain, snow, or fog. The chemical reactions involved include: $SO_2 + H_2O \rightarrow H_2SO_3$ (sulphurous acid) $SO_3 + H_2O \rightarrow H_2SO_4$ (sulphuric acid) $2NO_2 + H_2O \rightarrow HNO_2 + HNO_3$ Effects of acid rain include damage to buildings, soil acidification, harm to aquatic life, and negative impacts on plant growth and human health. It is a major environmental concern, especially in industrial and urban areas.		
3	**Explain BOD and COD as water quality parameters.**	**Level-2**	**CO4**
Ans.	Biological Oxygen Demand (BOD) and Chemical Oxygen Demand (COD) are important parameters used to assess the quality of water, especially in terms of organic pollution. BOD measures the amount of oxygen consumed by microorganisms to decompose organic matter in water over a specific time period, usually 5 days at 20°C. A high BOD value indicates a high level of organic pollution, which can reduce oxygen available to aquatic life. COD measures the total amount of oxygen required to chemically oxidize both organic and inorganic pollutants in water. It provides a quicker and broader estimate of water pollution compared to BOD. Both parameters are used to evaluate the effectiveness of wastewater treatment and to monitor pollution in water bodies. Normal water should have low BOD and COD values to support healthy aquatic ecosystems.		
4	**Mention any three effects of noise pollution on human health.**	**Level-2**	**CO4**

Ans.	Noise pollution has several harmful effects on human health, especially when exposure is prolonged or intense. Three major effects include: Hearing Loss: Continuous exposure to high noise levels (above 85 dB) can cause temporary or permanent hearing damage, including tinnitus and partial deafness. Psychological Stress: Noise pollution can lead to stress, anxiety, irritability, and sleep disturbances, affecting mental well-being and daily functioning. Cardiovascular Issues: Prolonged exposure to loud noise can raise blood pressure, cause irregular heartbeats, and increase the risk of heart-related diseases. These effects show that noise pollution is not just an annoyance but a serious public health concern that requires control and preventive measures.		
5	**List any three sources of marine pollution and briefly explain anyone.**	**Level-2**	**CO4**
Ans.	Three major sources of marine pollution are: Oil spills during transportation or storage Industrial waste discharge into oceans Plastic and solid waste dumping Explanation (Oil spills): Oil spills occur when petroleum is accidentally released into the sea during drilling, transportation, or storage. Oil floats on water, spreading quickly and forming a thin layer that blocks sunlight and oxygen exchange. This affects marine organisms, damages coastal ecosystems, and takes years to clean up, making it one of the most harmful forms of marine pollution.		
6	**Write a short note on vermicomposting.**	**Level-2**	**CO4**
Ans.	Vermicomposting (often misspelled as "vermin composting") is a biological process that uses specific species of earthworms, such as Eisenia fetida (red wigglers), to decompose organic waste into a nutrient-rich material called vermicompost. In this method, organic materials like vegetable scraps, paper, and plant waste are placed in a composting bin. Earthworms feed on this waste, and through digestion, convert it into castings—a fine, odorless, and dark compost that is rich in nitrogen, phosphorus, potassium, and other micronutrients. Benefits of vermicomposting include: Reducing household and agricultural organic waste Producing high-quality, natural fertilizer Improving soil structure and fertility		

	Promoting sustainable and eco-friendly waste management It is a simple and efficient technique suitable for both urban and rural areas.
PART–C 10 Marks Questions	

1	**Discuss in detail the classification of air pollutants and their effects on human health and environment.**	**Level-3**	**CO4**
Ans.	**Vermicomposting:** Vermicomposting is an eco-friendly method of managing organic waste using earthworms—particularly Eisenia fetida (red wigglers)—to decompose biodegradable materials into a nutrient-rich substance called vermicompost. This compost is an excellent organic fertilizer that improves soil fertility, aeration, and water retention capacity. During vermicomposting, earthworms consume organic waste such as kitchen scraps, paper, and garden waste. Inside their digestive systems, the material is broken down by enzymes and microbes. The waste is then excreted as castings, which are rich in essential plant nutrients like nitrogen (N), phosphorus (P), and potassium (K), along with beneficial microorganisms. Benefits of Vermicomposting: Converts waste into a natural fertilizer, reducing dependence on chemical fertilizers. Helps in solid waste management, especially in urban areas. Enhances soil structure and biological activity. Encourages sustainable agriculture and supports environmental conservation. 2. Classification of Air Pollutants and Their Effects on Human Health and the Environment: Air pollutants can be classified based on origin and physical state: A. Based on Origin Primary Pollutants These are pollutants directly emitted from a source into the atmosphere. Examples: Carbon monoxide (CO) Sulphur dioxide (SO_2) Nitrogen oxides (NO, NO_2) Hydrocarbons (CH_4, C_2H_6) Chlorofluorocarbons (CFCs)		
2	**Secondary Pollutants**	**Level-3**	**CO4**
Ans.	These are not emitted directly but form in the atmosphere through chemical reactions among		

	primary pollutants in the presence of sunlight. Examples: Ozone (O_3) Peroxyacetyl nitrate (PAN) Smog Acid rain (H_2SO_4, HNO_3) B. Based on Physical State Solid Pollutants: Dust, smoke, soot, and particulate matter like $PM_{2.5}$ and PM_{10} Liquid Pollutants: Mist, fog, and liquid aerosols Gaseous Pollutants: CO, NO_x, SO_2, VOCs (Volatile Organic Compounds) Effects on Human Health Carbon Monoxide (CO): Binds with hemoglobin in the blood, reducing oxygen transport, causing fatigue and even death at high levels. Sulphur Dioxide (SO_2): Causes respiratory problems, bronchitis, and eye irritation. Nitrogen Dioxide (NO_2): Irritates lungs, reduces lung function, and contributes to respiratory diseases. Ozone (O_3): Causes chest pain, coughing, and throat irritation; harmful to lungs and vegetation. Particulate Matter ($PM_{2.5}$ and PM_{10}): Penetrates deep into lungs and bloodstream, causing asthma, cardiovascular diseases, and lung cancer. Chlorofluorocarbons (CFCs): Although not directly toxic, they deplete the ozone layer, increasing risks of skin cancer and eye damage from UV radiation. Effects on Environment Acid Rain: Damages buildings, corrodes metals, acidifies soils and water bodies, and harms aquatic life. Smog: Reduces visibility and causes respiratory issues. Ozone Layer Depletion: Leads to increased UV radiation, harming crops, aquatic systems, and living organisms. Climate Change: Emissions like CO_2, CH_4, and NO_x act as greenhouse gases, contributing to global warming and climate disruptions.		
3	**Explain the process of eutrophication, its consequences, and**	**Level-3**	**CO4**

	measures to control it.		
Ans.	Eutrophication: Process, Consequences, and Control Measures Eutrophication is a process by which water bodies become overly enriched with nutrients, especially nitrogen (N) and phosphorus (P). This nutrient enrichment leads to excessive growth of algae and aquatic plants, a phenomenon commonly referred to as algal blooms or water bloom. Though eutrophication can occur naturally over centuries, human activities such as agricultural runoff, sewage discharge, and industrial effluents have greatly accelerated the process, causing serious environmental problems. Process of Eutrophication: Nutrients such as nitrates (NO_3^-) and phosphates (PO_4^{3-}) enter water bodies through agricultural runoff, untreated sewage, industrial waste, and detergents. These nutrients stimulate the rapid growth of algae, leading to dense algal blooms on the water surface. The algae eventually die and sink to the bottom. Microorganisms decompose the dead algae using up the dissolved oxygen (DO) in the water. As oxygen levels drop (a condition called hypoxia), aquatic organisms such as fish, invertebrates, and plankton struggle to survive or suffocate and die. Consequences of Eutrophication: Oxygen Depletion (Hypoxia): Leads to fish kills and disrupts the aquatic food chain. Loss of Biodiversity: Sensitive aquatic organisms die, allowing only tolerant species to survive. Algal Toxins: Some algae produce harmful toxins that can contaminate drinking water and affect human and animal health. Poor Water Quality: Water becomes turbid, foul-smelling, and unfit for consumption or recreational use. Disruption of Ecosystem Services: Affects fisheries, tourism, and the aesthetic value of water bodies. Measures to Control Eutrophication: Preventive Measures (Source Reduction): Reduce Fertilizer Use: Promote the use of biofertilizers and adopt precision farming to minimize runoff. Wastewater Treatment: Ensure sewage and industrial effluents are treated before being discharged into water bodies. Buffer Zones: Establish vegetative buffer strips along rivers and lakes to absorb nutrients.		

	Public Awareness: Educate people about the impacts of excessive detergent and fertilizer use. Remedial Measures (In-Process Control): Aeration of Water Bodies: Use mechanical aerators to increase DO levels and support aquatic life. Chemical Treatment: Use of chemicals like aluminum sulfate (alum) to precipitate phosphates from water. Algae Removal: Physically remove algae using skimmers or apply algaecides cautiously. Constructed Wetlands: Develop artificial wetlands to naturally filter out nutrients from incoming water.		
4	**Describe the sources, effects, and control measures of marine pollution.**	**Level-4**	**CO4**
Ans.	**1. Sources of Marine Pollution** Marine pollution refers to the introduction of harmful substances or pollutants into the ocean, which disrupts marine ecosystems and affects biodiversity. These pollutants primarily arise from the following sources: **a) Land-Based Sources** 1. **Agricultural Runoff**: Excessive use of fertilizers, pesticides, and herbicides washes into rivers, eventually reaching the oceans. These substances can cause eutrophication (excessive nutrients leading to algal blooms) and harm marine life. 2. **Industrial Waste**: Factories discharge harmful chemicals such as heavy metals, solvents, and toxins into rivers and oceans. 3. **Sewage and Wastewater**: Domestic and industrial wastewater, often untreated, contains bacteria, pathogens, and chemicals that can contaminate marine life and water. 4. **Plastic Waste**: Single-use plastics like bottles, bags, and packaging materials are major contributors to ocean pollution, harming marine organisms and entering the food chain. 5. **Oil Spills**: Accidental discharge of oil from ships or offshore drilling rigs can cause long-lasting harm to marine ecosystems and coastal environments. **b) Marine-Based Sources** 1. **Shipping and Marine Transport**: Ships release oil, garbage, and ballast water into the ocean. Ballast water often contains invasive species, which can disrupt local marine ecosystems. 2. **Fishing Activities**: Overfishing and the disposal of fishing nets, lines, and equipment contribute to marine pollution. Ghost nets (lost or discarded fishing nets) entangle marine		

life, causing injuries or death.

3. **Offshore Drilling**: Drilling for oil and gas releases chemicals and oil into the marine environment, which can cause devastating ecological damage.

2. Effects of Marine Pollution

Marine pollution has far-reaching and diverse impacts on the ocean's ecosystems and the organisms that depend on them:

a) Ecological Impacts

1. **Loss of Marine Biodiversity**: Pollution, especially plastic and chemical pollutants, can lead to the death of marine species, including fish, mammals, and birds. Coral reefs, which support a diverse range of species, are also severely affected by pollution.
2. **Disruption of the Food Chain**: Toxic substances and plastics enter the food chain through plankton, which are eaten by fish, and then consumed by larger predators, including humans. This leads to bioaccumulation and biomagnification of toxins.
3. **Eutrophication**: Excess nutrients from agricultural runoff lead to algal blooms, which deplete oxygen levels in the water, causing dead zones (areas with insufficient oxygen to support marine life).
4. **Acidification**: Carbon dioxide (CO_2) emissions from human activities are absorbed by the oceans, leading to ocean acidification. This harms organisms like corals and shellfish that rely on calcium carbonate to form shells and skeletons.

b) Economic Impacts

1. **Loss of Fisheries**: Pollution reduces fish populations and harms aquatic ecosystems, impacting the livelihoods of fishermen and the global seafood industry.
2. **Tourism Decline**: Polluted beaches and coastal waters discourage tourism, leading to economic losses for local economies reliant on marine tourism.
3. **Increased Cleanup Costs**: Oil spills and other forms of pollution require expensive cleanup operations, diverting government resources from other necessary programs.

c) Human Health Impacts

1. **Contamination of Seafood**: Pollutants like heavy metals and toxins accumulate in fish and shellfish, which, when consumed by humans, can lead to serious health issues such as mercury poisoning.
2. **Waterborne Diseases**: Sewage and untreated wastewater introduce pathogens into coastal waters, increasing the risk of waterborne diseases like cholera, dysentery, and hepatitis.

	3. Control Measures for Marine Pollution Addressing marine pollution requires global, regional, and local efforts aimed at reducing the release of pollutants into marine environments and protecting ocean ecosystems. Some of the key control measures include: **a) Prevention and Reduction at Source** 1. **Improved Waste Management**: Efficient solid waste collection and recycling programs can reduce the amount of plastic and garbage that enters the oceans. Encouraging businesses and consumers to reduce single-use plastics is critical. 2. **Sustainable Agricultural Practices**: The use of organic fertilizers, integrated pest management, and proper irrigation techniques can minimize the runoff of harmful chemicals into the ocean. 3. **Proper Wastewater Treatment**: Ensuring that industrial and domestic wastewater is treated before being released into rivers and oceans can significantly reduce the levels of harmful chemicals and pathogens. 4. **Ban or Regulation on Toxic Substances**: Governments should regulate and phase out the use of harmful chemicals, such as certain pesticides and fertilizers, to prevent contamination of marine environments. **b) Pollution Control Technologies** 1. **Oil Spill Cleanup**: Technologies such as booms, skimmers, and absorbent materials are used to contain and clean up oil spills. Bioremediation (using microorganisms to break down oil) is another technique. 2. **Plastic Waste Collection**: Initiatives like "The Ocean Cleanup" project aim to use large floating barriers to collect plastic waste in ocean gyres, preventing it from harming marine life. 3. **Advanced Filtration Systems**: Implementing advanced filtration systems in ships and industrial outlets can prevent harmful substances from entering the ocean. **c) International Agreements and Regulations** 1. **The MARPOL Convention**: The International Convention for the Prevention of Pollution from Ships (MARPOL) regulates discharges from ships and oil platforms. It has been successful in reducing the amount of oil and other pollutants released into the marine environment. 2. **The Kyoto Protocol and Paris Agreement**: These international agreements on climate change aim to reduce carbon emissions, which will help prevent ocean acidification and

	other climate-related marine issues. 3. **Marine Protected Areas (MPAs)**: Establishing MPAs can help conserve marine biodiversity and allow ecosystems to recover from pollution-related damage. **d) Public Awareness and Education** 1. **Community Engagement**: Educating the public about the importance of reducing plastic waste, responsible fishing, and protecting marine life can help change behaviors and reduce pollution at the grassroots level. 2. **Marine Cleanup Initiatives**: Organizing beach clean-ups and other community-based initiatives can reduce visible pollution and raise awareness about the impact of waste on marine ecosystems.

Unit V Question bank

Part--- A 2 Marks Questions			
1	What is sustainable development?	**Level-1**	CO5
Ans.	Development that meets the needs of the present without compromising the ability of future generations to meet their own needs. It aims to balance economic growth, environmental protection, and social equity.		
2	What are the salient features of wildlife protection act?	**Level-1**	CO5
Ans.	Salient Features of Wildlife Protection Act, 1972: 1. Provides legal protection to wild animals, birds, and plants. 2. Establishes Protected Areas like National Parks, Wildlife Sanctuaries, and Biosphere Reserves. 3. Regulates hunting and prohibits poaching and illegal trade of wildlife. 4. Empowers the central and state governments to take conservation measures. 5. Includes provisions for penalties and punishments for wildlife crimes.		
3	What is ecological footprint?	**Level-1**	CO5
Ans.	Measure of the amount of natural resources and land area required to support a person's or population's lifestyle. It assesses the environmental impact in terms of resource consumption and waste generation.		
4	What is green building concept?	**Level-2**	CO5
Ans.	Designing, constructing, and operating buildings in an environmentally responsible and resource-efficient way. It aims to reduce negative impacts on the environment by using energy, water, and materials efficiently.		
5	What are environmental ethics?	**Level-1**	CO5
Ans.	**Environmental ethics** is a branch of philosophy that studies the moral relationship between human beings and the natural environment. It deals with the ethical responsibilities humans have towards nature, including animals, plants, ecosystems, and future generations.		
6	What is National Environmental policy?	**Level-1**	CO5
Ans.	National Environment Policy (NEP) 2006 is a policy framework by the Government of India aimed at environmental protection and sustainable development.		
7	What are salient features of air act?	**Level-1**	CO5
Ans.	Salient Features of the Air (Prevention and Control of Pollution) Act, 1981: **Control of Air Pollution:** Aims to prevent, control, and reduce air pollution. **Pollution Control Boards:** Establishes Central and State Pollution Control Boards for		

	implementation. **Emission Standards:** Empowers boards to set standards for emissions from industries and vehicles.		
8	What are environmental ethics?	**Level-1**	CO5
Ans.	Study of relationship between humans and the environment. It promotes responsible use of natural resources and respect for all living beings, emphasizing the duty to protect and preserve nature for current and future generations.		
9	What is EIA?	**Level-1**	CO5
Ans.	EIA (Environmental Impact Assessment) is a process used to evaluate the environmental effects of a proposed project before it is approved. It helps in identifying, predicting, and mitigating potential negative impacts on the environment and promotes sustainable development.		
10	Define solid waste management?	**Level-1**	CO5
Ans.	Solid waste management refers to the collection, transportation, treatment, and disposal of solid waste in a safe and efficient manner. Its goal is to reduce the harmful effects of waste on human health and the environment.		
PART–B 5 Marks Questions			
1	Explain in detail water act?	**Level-2**	CO5
Ans.	The **Water (Prevention and Control of Pollution) Act, 1974** was enacted by the Government of India to prevent and control water pollution and to maintain or restore the wholesomeness of water. It was one of the first environmental laws in India to address water pollution comprehensively. **Objectives:** • To prevent and control water pollution. • To maintain or restore the quality of water bodies. • To establish pollution control boards for monitoring and enforcement. **Key Provisions:** 1. **Establishment of Pollution Control Boards**: The Act provides for the constitution of the **Central Pollution Control Board (CPCB)** and **State Pollution Control Boards (SPCBs)**. These bodies are responsible for planning and executing programs for the prevention and control of water pollution. 2. **Consent Mechanism**: Industries and other entities must obtain prior consent from		

	SPCBs before discharging sewage or trade effluents into water bodies. 3. **Monitoring and Inspection**: SPCBs are empowered to inspect treatment plants, collect samples, and take necessary legal actions against violators. 4. **Prohibition of Pollutant Discharge**: The Act prohibits the disposal of pollutants into streams, wells, and land without proper treatment. 5. **Penalties and Punishments**: The Act prescribes fines and imprisonment for non-compliance and pollution-related offenses. **Significance:** The Water Act plays a vital role in protecting India's water resources from industrial and domestic pollution. It promotes sustainable water use and holds polluters accountable, thereby safeguarding public health and aquatic ecosystems.		
2	Write salient features of air act?	**Level-2**	CO5
Ans.	**Salient Features of the Air (Prevention and Control of Pollution) Act, 1981** **Objective**: To prevent, control, and reduce air pollution and maintain air quality. 1. **Pollution Control Boards**: Establishes the **Central Pollution Control Board (CPCB)** and **State Pollution Control Boards (SPCBs)** for implementation and monitoring. 2. **Air Pollution Control Areas**: State governments, in consultation with SPCBs, can declare any area as an "air pollution control area" to regulate activities. 3. **Consent Requirement**: Industries must obtain permission from SPCBs before operating in air pollution control areas. 4. **Emission Standards**: CPCB sets emission standards for industries, vehicles, and other polluting sources. 5. **Powers of Boards**: Boards can inspect industrial units, collect air samples, and issue directions for pollution control.		

	6. **Prohibition of Pollutants**: Restricts emission of air pollutants beyond prescribed limits and bans the use of polluting fuels or substances. 7. **Penalties and Legal Action**: Provides for fines and imprisonment for violations, non-compliance, or obstruction of board officials. 8. **Coordination with Other Agencies**: Encourages coordination between central and state authorities for effective air quality management. 9. **Public Awareness and Research**: Promotes environmental awareness, technical research, and pollution control methods. The Air Act, 1981 is a critical legislative step toward ensuring clean air and controlling environmental degradation in India.		
3	Explain in detail wild life Act?	**Level-3**	CO5
Ans.	The **Wildlife (Protection) Act, 1972** is a comprehensive legislation enacted by the Government of India to safeguard the country's wildlife and their habitats. It was introduced in response to the increasing threats of poaching, habitat destruction, and illegal trade in wildlife. **Objectives:** • To protect wild animals, birds, and plants. • To establish protected areas for conservation. • To regulate hunting and ban illegal trade in wildlife and their products. **Key Provisions:** 1. **Protected Areas**: The Act provides for the establishment of **National Parks**, **Wildlife Sanctuaries**, **Conservation Reserves**, and **Community Reserves** to preserve biodiversity. 2. **Hunting Restrictions**: Hunting of wild animals is strictly prohibited except under specific conditions such as for scientific research or if an animal becomes dangerous to human life.		

	3. **Wildlife Crime Control**: The Act regulates trade and commerce in wild animals, their parts, and products. It includes strict provisions to control poaching and smuggling. 4. **Schedules of Protection**: The Act contains six schedules that classify species based on the level of protection they require. Schedule I and II species receive the highest protection. 5. **Penalties**: It includes stringent penalties for violations, including imprisonment and fines, especially for crimes involving species listed in Schedule I and II. 6. **Authorities**: The Act empowers both central and state governments to appoint forest officers and establish wildlife advisory boards for its implementation. **Importance:** The Wildlife Protection Act has played a crucial role in the recovery of endangered species and the conservation of India's rich biodiversity. It has helped in creating awareness and establishing a legal framework for wildlife conservation in the country.

4	Explain the important measures to be taken for sustainable development.	**Level-3**	CO5
Ans.	**Sustainable development** refers to development that meets the needs of the present without compromising the ability of future generations to meet their own needs. To achieve this balance between economic growth, environmental protection, and social equity, several important measures must be taken: **1. Conservation of Natural Resources** Sustainable use of resources like water, soil, forests, and minerals is essential. Overexploitation must be avoided, and renewable resources should be used efficiently. **2. Pollution Control** Efforts must be made to reduce air, water, and soil pollution through cleaner technologies,		

	proper waste management, and stricter pollution control laws. **3. Use of Renewable Energy** Replacing fossil fuels with renewable sources like solar, wind, hydro, and biomass reduces environmental degradation and supports long-term energy security. **4. Sustainable Agriculture** Practices like organic farming, crop rotation, and minimal use of chemical fertilizers ensure soil fertility and food security without harming the environment. **5. Eco-friendly Urban Planning** Cities should be planned with green spaces, public transport systems, and energy-efficient buildings to reduce environmental stress. **6. Environmental Education and Awareness** Educating people, especially youth, about environmental issues encourages responsible behavior and promotes conservation efforts. **7. Policy and Governance** Governments must implement and enforce environmental laws, create policies that promote sustainable industries, and support green technologies. **8. Inclusive Growth** Equitable access to resources, healthcare, and education ensures that economic benefits reach all sections of society, promoting social sustainability. In conclusion, sustainable development requires coordinated efforts at individual, community, national, and global levels to ensure a healthy planet for future generations.		
5	Mention the basic characteristics of green building	**Level-2**	CO5

Ans.	**Basic Characteristics of a Green Building** 1. **Energy Efficiency**: Uses energy-saving technologies like LED lighting, efficient HVAC systems, and renewable energy sources such as solar panels. 2. **Water Conservation**: Incorporates rainwater harvesting, low-flow fixtures, and wastewater recycling to minimize water use. 3. **Sustainable Materials**: Uses eco-friendly, recycled, or locally sourced construction materials to reduce environmental impact. 4. **Indoor Environmental Quality**: Ensures good ventilation, natural lighting, and non-toxic materials to promote occupant health and comfort. 5. **Waste Reduction**: Minimizes construction and operational waste through proper planning and recycling systems. 6. **Efficient Site Planning**: Preserves natural landscapes, maximizes green cover, and reduces the heat island effect through smart design. 7. **Reduced Carbon Footprint**: Aims to lower greenhouse gas emissions through energy and resource efficiency. 8. **Smart Design and Innovation**: Integrates modern technologies like automated systems and passive solar design for optimal performance.		
6	Discuss the salient features of ‘Forest conservation act’	**Level-2**	CO5
Ans.	The **Forest (Conservation) Act, 1980** was enacted by the Government of India to protect and conserve forests. It aims to regulate the diversion of forest land for non-forest purposes and ensure sustainable forest management. **Key Objectives:**		

	• To prevent deforestation and conserve forest resources. • To maintain ecological balance and protect biodiversity. • To regulate the use of forest land for development purposes. **Salient Features:** 1. **Restriction on Non-Forest Use**: Forest land cannot be diverted for non-forest purposes (such as agriculture, industry, or mining) without the prior approval of the **Central Government**. 2. **Centralized Control**: The Act centralizes decision-making on forest land diversion, limiting the powers of state governments to approve projects. 3. **Advisory Committee**: A central advisory committee is constituted to review proposals for forest land use and advise on conservation measures. 4. **Compensatory Afforestation**: If forest land is diverted, the user agency must undertake afforestation on an equivalent area of non-forest land, to compensate for the loss. 5. **Penalties for Violation**: Unauthorized use of forest land or violation of the Act's provisions can lead to legal action, including fines and imprisonment. 6. **Applicable to All Forests**: The Act covers all categories of forests, whether notified or not, and irrespective of ownership. **Conclusion:** The Forest (Conservation) Act, 1980 plays a critical role in regulating forest use, preventing deforestation, and promoting environmental sustainability in India. It ensures that development activities are balanced with ecological preservation.
PART–C 10 Marks Questions	

1	Discuss in detail environmental protection act?	**Level-3**	CO5

Ans.	The **Environmental Protection Act, 1986** was enacted by the Government of India in the wake of the Bhopal Gas Tragedy. It provides a comprehensive legal framework for protecting and improving the environment and empowers the central government to take necessary measures for environmental conservation. **Objectives:** • To provide for the protection and improvement of the environment. • To prevent hazards to human beings, living creatures, and property. • To empower the central government to regulate all forms of environmental pollution. **Salient Features:** 1. **Comprehensive Legislation**: The Act covers air, water, and land pollution and is considered an umbrella legislation for environmental protection in India. 2. **Central Government Powers**: The Act authorizes the central government to set environmental standards, regulate industrial locations, and control hazardous substances. 3. **Environmental Standards and Restrictions**: It allows for the laying down of emission and discharge standards for pollutants from various sources. 4. **Hazardous Substance Management**: The Act provides rules for handling, storage, and disposal of hazardous chemicals and wastes. 5. **Penalty Provisions**: Strict penalties are prescribed for violations, including imprisonment up to 5 years and/or fines. 6. **Right to Take Emergency Measures**: The government can take emergency action to prevent environmental damage or pollution. 7. **Public Participation**: Encourages citizen participation by allowing complaints against environmental violations.

	Importance: The Environmental Protection Act, 1986 acts as a backbone for all environmental laws in India. It gives broad powers to the government to protect the environment and serves as a foundation for various rules such as EIA, hazardous waste management, and noise pollution control.		
2	Write an essay on environmental impact assessments?	**Level-3**	CO5
Ans.	**Introduction:** Environmental Impact Assessment (EIA) is a crucial planning tool used to evaluate the potential environmental effects of a proposed development project before it is approved or implemented. It helps in identifying, predicting, and mitigating adverse impacts on the environment. **Purpose of EIA:** The primary aim of EIA is to ensure that decision-makers consider environmental concerns alongside economic and technical aspects. It promotes sustainable development by preventing environmental degradation before it occurs. **Process of EIA:** EIA follows a systematic process which includes the following steps: 1. **Screening** – to determine if a project requires EIA. 2. **Scoping** – to identify key issues and impacts. 3. **Baseline Study** – to assess the existing environmental conditions. 4. **Impact Prediction** – to evaluate possible environmental effects. 5. **Mitigation Measures** – to propose ways to reduce or avoid impacts. 6. **Public Consultation** – to involve stakeholders and communities. 7. **Reporting (EIA Report or Environmental Management Plan)** – to document findings. 8. **Decision-making and Monitoring** – to approve, reject, or suggest changes and monitor compliance. **Importance of EIA:** EIA helps in protecting natural resources, promoting public involvement, ensuring legal compliance, and improving project planning. It plays a vital role in balancing development needs with environmental conservation.		

<table>
<tr><td></td><td colspan="3">Examples of Projects Requiring EIA:
Large infrastructure projects such as highways, dams, power plants, mining operations, and industrial setups are typically subjected to EIA.</td></tr>
<tr><td>3</td><td>Discuss the scope and importance of EIA? How EIA is related to sustainable development?</td><td>Level-3</td><td>CO5</td></tr>
<tr><td>Ans.</td><td colspan="3">Environmental Impact Assessment (EIA) is a scientific and legal process used to evaluate the potential environmental effects of proposed projects before they are carried out. It plays a vital role in integrating environmental concerns into development planning and decision-making.

Scope of EIA:

1. Wide Application:
EIA applies to a variety of developmental activities such as highways, dams, mining, thermal and nuclear power plants, ports, and industries.
2. Comprehensive Assessment:
It covers physical (air, water, soil), biological (flora, fauna), and socio-economic (health, livelihood, resettlement) impacts of projects.
3. Preventive Tool:
EIA helps in forecasting and avoiding potential environmental damages before they occur.
4. Legal Requirement:
In many countries, including India, EIA is a mandatory requirement for major developmental projects under the Environmental Protection Act, 1986.
5. Public Involvement:
EIA includes stakeholder consultations, promoting transparency and community participation.

Importance of EIA:

• Protects Natural Resources: Prevents overexploitation and degradation of air, water, land, and biodiversity.
• Improves Project Design: Identifies alternatives and mitigation measures to minimize adverse effects.</td></tr>
</table>

- **Supports Decision-Making**: Provides a scientific basis for approving or rejecting projects.
- **Ensures Compliance**: Helps meet national environmental laws and international standards.
- **Reduces Long-Term Costs**: Prevents costly damage and legal complications by addressing risks early.

EIA and Sustainable Development:

EIA directly contributes to **sustainable development** by ensuring that economic growth does not come at the cost of environmental degradation. It balances developmental needs with ecological protection, helping meet the needs of the present without compromising the future. By promoting environmentally sound and socially responsible decisions, EIA acts as a bridge between development and sustainability.

4	**Discuss in detail solid waste management**	**Level-3**	**CO5**

Ans.

Solid waste management refers to the systematic handling of solid waste materials from their origin to final disposal. It includes the collection, segregation, transportation, treatment, and disposal of waste. Proper waste management is essential for protecting human health, conserving the environment, and promoting sustainable urban development.

Types of Solid Waste:

1. **Municipal Solid Waste (MSW):** Household and commercial waste like food scraps, paper, plastics, etc.
2. **Biomedical Waste:** Waste generated from hospitals, clinics, and labs (e.g., syringes, bandages).
3. **Industrial Waste:** Waste from manufacturing and industrial processes.
4. **Hazardous Waste:** Toxic, flammable, or corrosive waste requiring special handling.
5. **E-Waste:** Discarded electronic devices like computers, phones, etc.

Steps in Solid Waste Management:

1. **Waste Generation:** Originates from homes, institutions, industries, etc.

2. **Segregation:** Waste is separated into biodegradable, non-biodegradable, and hazardous categories at source.
3. **Collection and Storage:** Waste is collected in bins and transported by municipal services.
4. **Transportation:** Waste is transported to treatment or disposal sites.
5. **Processing/Treatment:** Includes composting, recycling, incineration, or energy recovery.
6. **Final Disposal:** Non-recyclable waste is disposed of in sanitary landfills or scientifically managed dumpsites.

Importance of Solid Waste Management:

- **Environmental Protection:** Reduces pollution of air, water, and soil.
- **Public Health:** Prevents the spread of diseases caused by improper waste disposal.
- **Resource Recovery:** Promotes recycling and reuse, conserving natural resources.
- **Aesthetic Value:** Maintains cleanliness and visual appeal of surroundings.
- **Supports Sustainable Development:** Minimizes environmental footprint and enhances quality of life.

Conclusion:
Solid waste management is a critical component of environmental health and urban planning. Public participation, strict regulations, and innovative technologies are essential for effective and sustainable waste management practices.

GURU NANAK INSTITUTIONS TECHNICAL CAMPUS
SCHOOL OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF HUMANITIES AND SCIENCES
Subject: Environmental Science

	Unit-1 Ecosystem
	Multiple Choice Questions (MCQs)
1.	The organisms which feed on dead organisms and waste of living organisms are called ( )
	a) producers b) carnivores c) herbivores d) decomposers
2.	The progressive accumulation of some non-biodegradable chemicals through the food Chains is known as ( )
	a)ecological balance b)biological magnification c) trophic structure d) bio-degradation
3.	The over nourished lakes with 'algal blooms' are called ( )
	a)eutrophication b) oligotrophic c) dystrophic d) meromictic
4.	The study of interactions between living organisms and environment is called as ( )
	a)ecosystems b) ecology c) phytogeography d) Phytosociology
5.	The term ecology was introduced by ( )
	a)Haekel b) Odum c) Tansely d) Ramdeo Mishra
6.	The sequence of eating and being eaten in an ecosystem is called as___( )
	a)food web b)food chain c)bio magnification d)bio-accumulation
7.	Biomagnifications of the pesticide in an food chain resulted in the thinning of shells in birds eggs. ( )
	a)methane b)ethane c)DDT d)urea
8.	Example of y channel energy flow ( )
	a)grazing food chain b)detritus food chain c)grazing and detritus food chain d)grazing nor detritus food chain
9.	Movement of nutrients in an ecosystem is cyclic while flow of energy is__( )
	a)bidirectional b)tri directional c)multi directional d)unidirectional
10.	Guano deposits on the coasts of Peru are rich in the nutrient ( )
	a)nitrogen b)calcium c)phosphorous d)sulphur
11.	The process of releases NH3 is called ( )
	a) ammonification b) nitrification c) sulphonation d) acidification
12.	In nitogen fixation ammonia is converted to nitrates and nitraites in presence of ( )
	a)azobacterium b)rhizobium c)lactic acid d)nitrifying bacteria
13.	Pyramid of biomass is ( )
	a)inverted b)upright c)inverted and upright d)inverted nor upright

14.	Nitrients like carbon,nitrogen,hydrogen etc are move in circular path through biotic and abiotic components. hence known as ( )
	a) bio accumulation b) bio magnification c)bio synthesis d) bio-geochemical cycle
15.	Which bacteria converts nitrogen into ammonia ( )
	a)azobacterium b)rhizobium c)lactic acid d)nitrifying bacteria
16	. Eutrophication is caused due to excess of______________
17	The graphical representation of the tropic structure and also tropic function is called ___________
18	The amount of living matter at each trophic level at a given time is known as_______
19	The number of individuals of a given species that can be sustained indefinitely in a given space is known as ____________
20	In nitogen fixation ammonia is converted to nitrates and nitraites in presence of _____
21	__________presence of sunlight using simple inorganic substances and built up of complex substances
22	__________are the carnivores like lion, tiger etc. that eat secondary consumers
23	The process of releases NH3 is called ______
24	Bioaccumulation is the process by which _______accumulate in the tissues of living organisms over time.
25	Most of the carbon dioxide enters the living world through___________
26	Chemosynthetic organisms can produce organic matter through oxidation of____________in the absence of sun light.
27	The environment which has been modified by human activities is called _____
28	__________is complex, interconnected network of food chains within an ecosystem.
29	Pyramid of_____________is always upright.
30	__________feed on both plants and animals.

	Unit-II Natural Resources
	Multiple Choice Questions (MCQs)
1.	________Resources are inexhaustible resources which can be generated within a given span of time. ( )
	a) renewable b) nonrenewable c) non exhaustible d) inexhaustible
2.	_________resources cannot be generated. ( )
	a) renewable b) nonrenewable c) non exhaustible d) inexhaustible
3.	Plants use _____gas for photosynthesis. ( )

	a) SO_2 B) CO_2 C) NO_2 D) O_2
4.	Deforestation means _______ of forests. ( )
	a) planting b)cutting c) dwelling d) falling
5.	____ of geographical area of a country should be forest area. ( )
	a) 44% b) 66% c) 33% d) 99%
6.	Maximum number of dams in India is in the ______ state. ( )
	a) Delhi b) Uttar Pradesh c) goa d) Maharashtra
7.	_______dam is the highest dam on river Bhagirathi in Uttaranchal ( )
	a) Bakanangal b) Krishna c) Tehri d) Nagarjuna sager
8.	The _______dam on river Sutlej in H.P. is the largest dam in terms of capacity. ( )
	a) Bhakra b) Krishna c) Tehri d) Nagarjuna sager
9.	Ecological issue related with Tehri Dam are taken up by the leader of_______ Chipko movement. ( )
	a) Sh. Sundarlal Bahuguna b) Nehru c) Gandhi d) Ambedkar
10.	Environmental activist Medha Patkar has taken up issues related to _____Dam. ( )
	a) Krishna b) Godavari c) Narmada d) Kaveri
11.	About _____ of the earth's surface is covered by water. ( )
	a) 44% b) 66% c) 33% d) 97%
12.	Only ______of total water on earth is readily available to us in the form of groundwater and fresh water. ( )
	a)4% b) 6% c) 3% d) 9%
13.	A layer of sediment or rock that is highly permeable and contains water (ground water) is called an ( )
	a) Aquifer b) unconfined aquifer c) confined aquifer d) amplifier
14.	Aquifers which are overlaid by permeable earth material and are recharged by seeping water are called__ aquifer. ( )
	a) Aquifer b) unconfined aquifer c) confined aquifer d) amplifier
15.	Aquifers which are sandwiched between two impermeable layers of rocks or sediments are called __aquifers. ( )
	Fill in the Blanks
16	Resources that are derived from living organisms are called ___________ resources.
17	Overuse of ___________ resources can lead to water scarcity and droughts.
18	Dams are beneficial for ___________ but can also cause displacement of communities.
19	Renewable energy sources, such as ___________ and wind, are sustainable for the environment

20	Excessive ___________ can lead to soil erosion and loss of biodiversity.
21	___________ energy is a non-renewable source that contributes to pollution when burned.
22	When groundwater is used faster than it is replenished, it can lead to ___________ of water.
23	_________ resources, such as oil and natural gas, are finite and will eventually run out.
24	Dams help in preventing ___________ by controlling the flow of rivers.
25	___________ is a significant environmental issue caused by deforestation and unsustainable land use.
26	Non renewable energy resourse examples------
27	Renewable energy resourses examples----------------
28	An alternative energy source that reduces greenhouse gas emissions is----------
29	Excessive mining of mineral resources can lead to----------------
30	The main cause of floods is typically due to-------------------

	Unit-III Biodiversity
	Multiple Choice Questions (MCQs)
1	Morphine is derived from ( )
	a)Poppy b) cinchona c) acterium d) periwinkle
2	The drug extracted from mint is called ( )
	a)quinine b) paracetamol c) menthol d)amoldipine
3	The number of species present in a system is called ( )
	a) species diversity b)genetic diversity c)ecosystem diversity d) species and genetic diversity
4	Diversity indicates ( )
	a) poverty b) richness c)affluence d)infertility
5	Cause of loss of biodivesrsity ( )
	a) Habitat degradation and loss b) Invasion of non native species c) wet lands d) planting
6	Consumptive use value and Productive use value are ( )
	a) indirect_values b)optional values c)social values d)Direct_values
7	Sunderbans are examples of ( )

	a)biosphere reserves b)wild life sanctuaries c)national parks d) kamala nehru park
8	Which diversity refers to the variation of genes within species ( )
	a) Species diversity b)genetic diversity c)ecosystem diversity d) biodiversity
9	What are the values are associated with the social life, religion and ( )
	a) consumptive values b)optional values c) social values d)ethical values
10	quinine is from the brak of ( )
	a) cincona tree b)poppy c) acterium d) periwinkle
11	Illegal killing of prohibited endangered animals is called ( )
	a)poaching b)habitat loss c)man wildlife conflicts d) legal hunting
12	Which country has the largest number of flowering plants ( )
	a)India b) China c)Australia d) Brazil
13	Western ghats are very rich in endemic species of ( )
	a) birds b) lion c) amphibians d) turtles
14	Example of ex-situ conservation ( )
	a)Biosphere reserve b) Gene bank c) Sanctuary d) national parks
15	What are the values involves like "all life must be preserved".
	a) ethical values b)optional values c)consumptive values d) social values
16	Sariska National Park is famous for ___________________
17	Ghana bird sanctuary is famous for _________________________
18	Bandipur national park located in __________________________
19	Example of in situ conservation ______________________________
20	Which region is known for having the highest biodiversity in the world ______________
21	The term for the variety of genes within a species__________
22	What is the primary cause of the current high rates of species extinction_____________
23	Indo-burma is rich in endemic species of ____________
24	Shannon-winer index gives a measure of _____________ diversity
25	Red data book giving the list of endangered species of plants and animals is published by _______
26	Cryopreservation of plant seeds and pollen is done at avery low temperature of -196c by using _______________

27	Sailent valley located at ___________________
28	The India's First National Marine Park ______________
29	An ecological state where in a species is introduced to a location where they are unique. ____________________
30	When the last individual in a particular species dies, it is known as ___________.

	Unit-IV Environmental Pollution
S. No.	**Multiple Choice Questions (MCQs)**
1	Which of the following is a secondary pollutant? ( )
	a) Carbon monoxide b) Sulphur dioxide b) Ozone b) Methane
2	Which air pollutant binds with haemoglobin and reduces oxygen transport in the blood ( )
	a) NO b) CO b) SO b) O
3	Which gas is primarily responsible for photochemical smog? ( )
	a) Nitrous oxide b) CO b) Hydrocarbons b) Methane
4	What does COD measure in water quality analysis? ( )
	a) Oxygen required by fish b) Toxicity of metals b) Total oxygen needed to oxidize pollutants b) Nutrient content
5	Which of the following is NOT a heavy metal pollutant? ( )
	a) Lead b) Mercury b) Arsenic b) Nitrogen
6	Which disease is caused by nitrate pollution in infants? ( )
	a) Cholera b) Methemoglobinemia b) Typhoid b) Dysentery
7	What is the threshold of pain for noise level? ( )
	a) 60 dB b) 85 dB b) 100 dB b) 120 dB
8	Which technique is used to remove suspended particulates from flue gases? ( )
	a) Scrubber b) Electrostatic Precipitator b) Filtration b) Sedimentation
9	Which type of agriculture input primarily leads to soil salinization? ( )
	a) Pesticides b) Over-irrigation b) Fertilizers b) Herbicides
10	What is the main component of domestic solid waste? ( )
	a) Heavy metals b) Plastics and organic matter b) Industrial solvents b) E-waste

11	11. Which of the following is a valuable metal found in e-waste? ( )
	a) Uranium b) Copper b) Zinc b) Nickel
12	Which of the following is a characteristic of e-waste? ( )
	a) Easily biodegradable b) High moisture content b) High metal content b) Radioactive decay
13	Which of the following is a greenhouse gas? ( )
	a) CO b) CO b) SO b) H
14	Which protocol aims to reduce ozone-depleting substances? ( )
	a) Kyoto Protocol b) Montreal Protocol b) Basel Convention b) Stockholm Convention
15	Which international agreement deals with climate change and GHG reduction? ( )
	a) Earth Charter b) Montreal Protocol b) Kyoto Protocol b) Paris Agreement
	Fill in the blanks
16	Pollution is defined as an undesirable change in the ______, ______, or ______ characteristics of the environment.
17	______ pollutants are emitted directly into the atmosphere from identifiable sources.
18	______ is formed by the reaction of SO_2 and NO_x with water vapour in the atmosphere.
19	The pollutant ______ is mainly responsible for the depletion of the ozone layer.
20	The optimal pH range for safe drinking water is between ______ and ______.
21	BOD stands for ______.
22	The disease ‘Minamata’ is caused by poisoning due to ______.
23	Excessive nitrates in drinking water can cause ______ in infants.
24	______ refers to the unwanted or harmful sound in the environment.
25	Hearing damage starts to occur at sound levels above ______ dB.
26	______ pollution refers to contamination of the soil with toxic substances.
27	One major source of soil pollution due to electronic goods is ______.
28	The process of using earthworms to convert waste into compost is called ______.
29	The three R’s of solid waste management are ______, ______, and ______.
30	The Montreal Protocol was signed to eliminate substances responsible for ______.

	UNIT V: Environmental Policy, Legislation and EIA
	Multiple Choice Questions (MCQs)
1.	Under which act is the Central Pollution Control Board constituted? ( )

	a)Air Act, 1981 b) Water Act, 1974 c) Environment Protection Act, 1986 d) Forest Conservation Act, 1980
2.	Which year was the Environment Protection Act enacted? ( )
	a)1972 b) 1980 c) 1981 d) 1986
3.	The Forest Conservation Act aims to prevent: ( )
	a)Mining in urban areas b) Deforestation and diversion of forest land c) Air pollution in forests d) Use of plastic in forests
4.	What is the main objective of the Air Act, 1981? ( )
	a)Control of soil erosion b) Conservation of wetlands c) Prevention and control of air pollution d) Preservation of biodiversity
5.	Biomedical Waste Management Rules were introduced in ( )
	a)1998 b) 2016 c) 2000 d) 2005
6.	What is the primary purpose of an Environmental Impact Assessment (EIA)? ( )
	a)Tree plantation b) Assessment of project cost c) Predicting and evaluating environmental impacts of a project d) Industrial licensing
7.	Which of the following is NOT a method of baseline data acquisition in EIA? ( )
	a)Field surveys b) Satellite imagery c) Financial audits d) Laboratory analysis
8.	Which of the following is an air quality parameter assessed in EIA? **( )**
	a)BOD b) SO_2 c) pH d) Turbidity
9.	Environmental Management Plan (EMP) is developed to **( )**
	a)Stop construction projects b) Increase industrial waste c) Mitigate environmental risks d) Promote consumerism
10.	What does Sustainable Development focus on? **( )**
	a)Only economic growth b) Depletion of natural resources c) Meeting present needs without compromising future generations d) Increasing urban sprawl
11.	What is the main problem of "Crazy Consumerism"? **( )**
	a)High taxation b) Environmental degradation due to over-consumption c) Rapid industrialization d) Rural development
12.	What is the purpose of Environmental Education? **( )**
	a)Promote tourism b) Increase consumerism c) Create awareness and responsibility towards the environment d) Encourage deforestation
13.	Urban Sprawl refers to: **( )**
	a)Sustainable city growth b) Dense forest growth in urban areas c) Unplanned and haphazard expansion of cities d) Reduction in urban populations
14.	The main focus of Life Cycle Assessment (LCA)is: **( )**
	a)Education cycle b) Energy consumption only c) Assessing environmental impacts throughout a product’s life d) Water conservation
15.	A Green Building aims to: **( )**
	a)Increase electricity use b) Reduce environmental impacts through sustainable design c) Expand construction with no regulation d) Use only non-recyclable materials

	Fill in the Blanks
16	The Environment Protection Act was enacted in the year ________.
17	The Water Act was passed in the year ________.
18	The Air (Prevention and Control of Pollution) Act was enacted in ________.
19	Forest Conservation Act was introduced in the year ________.
20	The Wildlife Protection Act came into force in the year ________.
21	________ is the central authority responsible for implementing environmental laws in India
22	The primary aim of EIA is to assess ________ impacts of development projects.
23	SO_2 and NOx are major pollutants in ________ pollution.
24	BOD and COD are important indicators of ________ pollution.
25	The concept of ________ ensures meeting the needs of the present without compromising future generations.
26	The impact of human activity on natural resources is measured through ________.
27	The process of examining the environmental impacts of a product from cradle to grave is called ________.
28	________ is the study of ethical relationship between humans and the environment.
29	The uncontrolled and unplanned growth of cities is known as ________.
30	The main aim of a ________ building is to minimize energy consumption and waste.

Unit-1 key		**Unit- 2 key**		**Unit-3 key**		**Unit-4 key**		**Unit-5 key**	
1	d	1	a	1	a	1	c	1	b
2	b	2	b	2	c	2	b	2	d
3	a	3	b	3	a	3	c	3	b
4	b	4	b	4	b	4	c	4	c
5	a	5	c	5	a	5	d	5	b
6	b	6	d	6	d	6	b	6	c
7	c	7	c	7	a	7	d	7	c
8	c	8	a	8	b	8	b	8	b
9	d	9	a	9	c	9	b	9	c
10	c	10	c	10	a	10	b	10	c
11	a	11	d	11	a	11	b	11	b
12	d	12	c	12	d	12	c	12	c

13	c	13	a	13	c	13	b	13	c
14	d	14	b	14	b	14	b	14	c
15	b	15	c	15	a	15	c	15	b
16	Phosphorous	16	Biotic resources.	16	Tiger	16	physical, chemical, biological	16	1986
17	ecological pyramids	17	Water	17	Bird sancturies	17	Primary	17	1974
18	standing crop	18	Water storage	18	Karnataka	18	Acid rain	18	1981
19	carrying capacity	19	Solar	19	Biospheres	19	Chlorofluorocarbons (CFCs)	19	1980
20	nitrifying bacteria	20	Use of fertilizers	20	Amazon forest	20	6.5, 8.5	20	1972
21	Autotrophs	21	Fossil fuels	21	Genetic diversity	21	Biological Oxygen Demand	21	Central Pollution Control Board environmental laws in India.
22	tertiary concumers	22	Over utilization	22	Human activites	22	Mercury (Hg)	22	Environmental development projects.
23	Ammonification	23	Non renewable	23	Fresh water turtiles	23	Methemoglobinemia (blue baby syndrome)	23	air
24	Toxic substances	24	Floods	24	Species	24	Noise pollution	24	water
25	Photo synthesis	25	Climate change.	25	IUCN	25	85	25	sustainable development
26	Inorganic compounds	26	Coal ,petroleum	26	Liquid Nitrogen	26	Soil	26	ecological footprint
27	Anthropogenic environment	27	Tidalenergy, wind energy	27	Kerala	27	E-waste	27	Life Cycle Assessment (LCA)
28	Food web	28	Solar energy	28	Gulf of kutch	28	Vermicomposting	28	Environmental Ethics
29	Energy	29	Soil erosion	29	Endemic species	29	Reduce, Reuse,Recycle	29	Urban Sprawl
30	Omnivores	30	Heavy rains and poor drainage systems	30	Extinction	30	Ozone layer depletion	30	Green